

Making Everything Easier!™

Learn to:

- **Recognize the benefits and risks of cloud services**
- **Understand the business impact and the economics of the cloud**
- **Govern and manage your cloud environment**
- **Develop your cloud services strategy**

Judith Hurwitz

Robin Bloor

Marcia Kaufman

Fern Halper

Get More and Do More at Dummies.com® Start with **FREE** Cheat Sheets

Cheat Sheets include

- Checklists
- Charts
- Common Instructions
- And Other Good Stuff!

**To access the Cheat Sheet created specifically for this book,
go to
*www.dummies.com/cheatsheet/cloudcomputing***

Get Smart at Dummies.com

Dummies.com makes your life easier with 1,000s of answers on everything from removing wallpaper to using the latest version of Windows.

Check out our

- Videos
- Illustrated Articles
- Step-by-Step Instructions

Plus, each month you can win valuable prizes by entering our Dummies.com sweepstakes. *

Want a weekly dose of Dummies? Sign up for Newsletters on

- Digital Photography
- Microsoft Windows & Office
- Personal Finance & Investing
- Health & Wellness
- Computing, iPods & Cell Phones
- eBay
- Internet
- Food, Home & Garden

Find out “HOW” at Dummies.com

**Sweepstakes not currently available in all countries; visit Dummies.com for official rules.*

**by Judith Hurwitz, Robin Bloor,
Marcia Kaufman, and Dr. Fern**

Halper

Cloud Computing For Dummies®

Published by

Wiley Publishing, Inc.

111 River Street

Hoboken, NJ 07030-5774

www.wiley.com

Copyright © 2010 by Wiley Publishing, Inc., Indianapolis, Indiana

Published by Wiley Publishing, Inc., Indianapolis, Indiana

Published simultaneously in Canada

No part of this publication may be reproduced, stored in a retrieval system or transmitted in any form or by any means, electronic, mechanical, photocopying, recording, scanning or otherwise, except as permitted under Sections 107 or 108 of the 1976 United States Copyright Act, without either the prior written permission of the Publisher, or authorization through payment of the appropriate per-copy fee to the Copyright Clearance Center, 222 Rosewood Drive, Danvers, MA 01923, (978) 750-8400, fax (978) 646-8600. Requests to the Publisher for permission should be addressed to the Permissions Department, John Wiley & Sons, Inc., 111 River Street, Hoboken, NJ 07030, (201) 748-6011, fax (201) 748-6008, or online at <http://www.wiley.com/go/permissions>.

Trademarks: Wiley, the Wiley Publishing logo, For Dummies, the Dummies Man logo, A Reference for the Rest of Us!, The Dummies Way, Dummies Daily, The Fun and Easy Way, Dummies.com, Making Everything Easier, and related trade dress are trademarks or registered trademarks of John Wiley & Sons, Inc. and/or its affiliates in the United States and other countries, and may not be used without written permission. All other trademarks are the property of their respective owners. Wiley Publishing, Inc., is not associated with any product or vendor mentioned in this book.

LIMIT OF LIABILITY/DISCLAIMER OF WARRANTY: THE PUBLISHER AND THE AUTHOR MAKE NO REPRESENTATIONS OR WARRANTIES WITH RESPECT TO THE ACCURACY OR COMPLETENESS OF THE CONTENTS OF THIS WORK AND SPECIFICALLY DISCLAIM ALL WARRANTIES, INCLUDING WITHOUT LIMITATION WARRANTIES OF FITNESS FOR A PARTICULAR PURPOSE. NO WARRANTY MAY BE CREATED OR EXTENDED BY SALES OR PROMOTIONAL MATERIALS. THE ADVICE AND STRATEGIES CONTAINED HEREIN MAY NOT BE SUITABLE FOR EVERY SITUATION. THIS WORK IS SOLD WITH THE UNDERSTANDING THAT THE PUBLISHER IS NOT ENGAGED IN RENDERING LEGAL, ACCOUNTING, OR OTHER PROFESSIONAL SERVICES. IF PROFESSIONAL ASSISTANCE IS REQUIRED, THE SERVICES OF A COMPETENT PROFESSIONAL PERSON SHOULD BE SOUGHT. NEITHER THE PUBLISHER NOR THE AUTHOR SHALL BE LIABLE FOR DAMAGES ARISING HEREFROM. THE FACT THAT AN ORGANIZATION OR WEBSITE IS REFERRED TO IN THIS WORK AS A CITATION AND/OR A POTENTIAL SOURCE OF FURTHER INFORMATION DOES NOT MEAN THAT THE AUTHOR OR THE PUBLISHER ENDORSES THE INFORMATION THE ORGANIZATION OR WEBSITE MAY PROVIDE OR RECOMMENDATIONS IT MAY MAKE. FURTHER, READERS SHOULD BE AWARE THAT INTERNET WEBSITES LISTED IN THIS WORK MAY HAVE CHANGED OR DISAPPEARED BETWEEN WHEN THIS WORK WAS WRITTEN AND WHEN IT IS READ.

For general information on our other products and services, please contact our Customer Care Department within the U.S. at 877-762-2974, outside the U.S. at 317-572-3993, or fax 317-572-4002.

For technical support, please visit www.wiley.com/techsupport.

Wiley also publishes its books in a variety of electronic formats. Some content that appears in print may not be available in electronic books.

Library of Congress Control Number: 2009938254

ISBN: 978-0-470-48470-8

Manufactured in the United States of America

10 9 8 7 6 5 4 3 2 1

About the Authors

Judith Hurwitz is a technology strategist and thought leader. She is the president of Hurwitz & Associates, a business technology strategy firm that helps companies gain business benefit from their technology investments. Her area of focus is on cloud computing and all the related distributed computing

technologies that enable the cloud. In 1992, she founded the Hurwitz Group, a technology research group. She has worked in various corporations, such as John Hancock, Apollo Computer, and Patricia Seybold's Group. She publishes a regular blog. Judith holds a BS and an MS degree from Boston University. She is a coauthor of *Service Oriented Architecture For Dummies*, Second Edition (Wiley), *Information on Demand For Dummies* (2009), *Service Management For Dummies* (2009), and *Collaboration For Dummies* (2009).

Robin Bloor, a partner with Hurwitz & Associates, has been an IT consultant and technology analyst for almost 20 years. He lived and worked in the U.K. until 2002, founding the IT analysis company Bloor Research, which published comparative technology reports that covered everything from computer hardware architecture to e-commerce. Robin is the author of the U.K. business bestseller, *The Electronic Bazaar: From the Silk Road to the E-Road* (Nicholas Brealey Publishing), which analyzed and explained the field of e-commerce. He is a coauthor of *Service Oriented Architecture For Dummies*, Second Edition (Wiley) and *Service Management For Dummies* (2009).

Marcia Kaufman, a founding partner of Hurwitz & Associates, has 20 years of experience in business strategy, industry research, and analytics. She has written many industry white papers and publishes a regular technology blog. Marcia has worked extensively on financial services industry modeling and forecasting in various research environments, including Data Resources, Inc. (DRI). Marcia holds an AB from Connecticut College in mathematics and economics and an MBA from Boston University. Marcia is coauthor of *Service Oriented Architecture For Dummies*, Second Edition (Wiley), *Information on Demand For Dummies* (2009), *Service Management For Dummies* (2009), and *Collaboration For Dummies* (2009).

Dr. Fern Halper, a partner with Hurwitz & Associates, has over 20 years of experience in data analysis, business analysis, and strategy development. Fern has published numerous articles on data and content management. She has done extensive research, writing, and speaking on the topic of text analytics. She publishes a regular technology blog. She has held key positions at AT&T Bell Laboratories and Lucent Technologies and directed strategy and product line planning for Lucent's Internet Software Unit. Fern received her BA from Colgate University and her Ph.D. from Texas A&M University. Fern is coauthor of *Service Oriented Architecture For Dummies*, Second Edition (Wiley), *Information on Demand For Dummies* (2009), and *Service Management For Dummies* (2009).

Authors' Acknowledgments

Amazon's Jeff Barr; Bell Aliant's Tony Lodge; CA's Brian Bonazzoli, Nicole Buffalino, Debra Cattani, Stephen Elliott, Jay Fry, Ajei Gopal, Joanne Moretti, Roger Pilc, and John Swainson; Cisco's William Scott; Citrix's Ian Platt; Cloud Camp's David Nielson; Computer Sciences Corporation's (CSC) Brian Boruff; Deskstone's Jeff Fisher and Harry Ruda; Distributed Management Task Force's (DMTF) Winston Bumpus; EMC's Chuck Hollis and Irene Mirageas; Good Data's Roman Stanek; GSK Pharmaceuticals' Ivan Hilaire; HP's Magdy Assem, Russ Daniels, Cheryl Rose Hayden, Tom Hogan, Rebecca Lawson, Scott McClellan, Joanne McMenoman, and Scott Pace; IBM's Lee Ackerman, Ruthie Amaru, Erich Clementi, Latha Colby, Teresa Cook, Jim Corgel, Dave Dworkin, Leon Katznelson, Martha Leversuch, Dave Lindquist, Amy Loomis, Steve Maher, Mike McCarthy, David Mitchell, Harold Moss, David Parker, Hamid Pirahesh, Sean Poulley, John Simonds, Toby Sirota, Zarina Lam Stanford, Lauren States, Tim Vincent, Marie Weeks, and David Yockelson; Intuit's Anna

Lane, Bill Lucchini, and Angus Thomson; JBoss's Aaron Darcy; MDot's Mike Kavis; Metro Health's Bill Lewkowski; Microsoft's Prashaut Ketkar, Niraj Nagrani, Steve Sloan, and Mike Warner; National Institute of Standards and Technology; Pervasive's John Bernard, Kimberli Daugherty, David Inbar, Jim Falgout, and Hollis Tibbetts; Platform Computing's Randy Clark; RightScale's Michael Crandell; Salesforce's Marc Benioff, Alex Chris, Ariel Kelman, and Bill Lukini; Savvis's Bryan Doerr; ServiceNow's Rhett Glauser; Sisters of Mercy Health System's Jeff Bell and John Treadway; State Street Corporation's David Saul; THINKStrategies's Jeff Kaplan; Virtual Bridges's Jim Curtin and Dan Perlman; VMware's Dawn Giusti, Neena Joshi, Wendy Perilli, and Jiam Zhen; Verizon's Joe Crawford and Tim Gillen; Wavemaker's Chris Keene; WorkXpress's Treff LaPlante; and 3tera's Paul Brennan.

Publisher's Acknowledgments

We're proud of this book; please send us your comments at <http://dummies.custhelp.com>. For other comments, please contact our Customer Care Department within the U.S. at 877-762-2974, outside the U.S. at 317-572-3993, or fax 317-572-4002.

Some of the people who helped bring this book to market include the following:
(www.the5thwave.com)

Acquisitions and Editorial

- Project Editor:** Tonya Maddox Cupp
- Development Editor:** Linda Morris,
Tonya Maddox Cupp
- Senior Acquisitions Editor:** Katie Feltman
- Technical Editor:** Brenda Michelson
- Editorial Manager:** Jodi Jensen
- Editorial Assistant:** Amanda Graham
- Sr. Editorial Assistant:** Cherie Case
- Cartoons:** Rich Tennant

Publishing and Editorial for Technology Dummies

- Composition Services**
- Project Coordinator:** Patrick Redmond
- Layout and Graphics:** Ashley Chamberlain,
Joyce Haughey, Christine Williams
- Proofreaders:** John Greenough,
Bonnie Mikkelsen
- Indexer:** Sharon Shock

- Richard Swadley**, Vice President and Executive Group Publisher
- Andy Cummings**, Vice President and Publisher
- Mary Bednarek**, Executive Acquisitions Director
- Mary C. Corder**, Editorial Director

Publishing for Consumer Dummies

- Diane Graves Steele**, Vice President and Publisher
- Composition Services**
- Debbie Stailey**, Director of Composition Services

Contents at a Glance

Introduction

1 Part I: Introducing Cloud

Computing.....	5
Chapter 1: Grasping the Fundamentals	7
Chapter 2: Discovering the Value of the Cloud for Business	17
Chapter 3: Getting Inside the Cloud	27
Chapter 4: Developing Your Cloud Strategy.....	39

Part II: Understanding the Nature of the Cloud.....

47 Chapter 5: Seeing the Advantages of the Highly Scaled Data Center	49
Chapter 6: Exploring the Technical Foundation for Scaling Computer Systems.....	59
Chapter 7: Checking the Cloud's Workload Strategy	67
Chapter 8: Managing Data	75
Chapter 9: Discovering Private and Hybrid Clouds.....	87

Part III: Examining the Cloud Elements

105 Chapter 10: Seeing Infrastructure as a Service	107
Chapter 11: Exploring Platform as a Service.....	119
Chapter 12: Using Software as a Service.....	137
Chapter 13: Understanding Massively Scaled Applications and Business Processes	153
Chapter 14: Setting Some Standards	161

Part IV: Managing the Cloud

171 Chapter 15: Managing and Securing Cloud Services	173
Chapter 16: Governing the Cloud	187
Chapter 17: Virtualization and the Cloud.....	197
Chapter 18: Managing Desktops and Devices in the Cloud.....	209
Chapter 19: Service Oriented Architecture and the Cloud.....	221
Chapter 20: Managing the Cloud Environment.....	231

Part V: Planning for the Cloud.....

243 Chapter 21: Banking on Cloud Economics	245
Chapter 22: Starting Your Journey to the Cloud	255

Part VI: The Part of Tens.....

265 Chapter 23: Ten (Plus One) Swell Cloud Computing Resources	267
Chapter 24: Ten Cloud Dos and	

Don'ts.....	271	Glossary.....	275
-------------	-----	---------------	-----

Index	291
-------------	-----

Table of Contents

Introduction	1
About	2
Assumptions.....	2
This Book Is Organized.....	2
Part I: Introducing Cloud Computing	3
Part II: Understanding the Nature of the Cloud.....	3
Part III: Examining the Cloud Elements.....	3
Part IV: Managing the Cloud.....	3
Part V: Planning for the Cloud.....	3
Part VI: The Part of Tens.....	3
Icons Used in This Book	4
Where to Go from Here.....	4

Part I: Introducing Cloud Computing.....	5
--	---

Chapter 1: Grasping the Fundamentals	7
--	---

Considering Perspectives.....	8
Computing on the Cloud.....	8
Cloud.....	9
Elasticity and scalability.....	10
Self-service provisioning.....	10
Application programming interfaces (APIs).....	11
Billing and metering of services.....	11
Performance monitoring and measuring.....	12
Security	12
Comparing Cloud Providers with Traditional IT Service Providers	12
Addressing Problems.....	13
Discovering the Business Drivers for Consuming Cloud Services	14
Supporting business agility	15
Reducing capital expenditures.....	15

Chapter 2: Discovering the Value of the Cloud for Business	17
--	----

Modeling

Services.....	17
Understanding Infrastructure as a Service.....	18
Exploring Platform as a Service.....	20
Seeing Software as a Service	21
Software as a Service modes	22
Massively scaled Software as a Service	23
Economies of scale	23
Management and Administration.....	24

xiv Cloud Computing For Dummies

Chapter 3: Getting Inside the Cloud 27

Feeling Sensational about Organization	27
Deciding on a strategy.....	28
Coping with governance issues	28
Monitoring business processes	29
Managing IT costs.....	30
Administering Cloud Services.....	30
Service level agreements and monitoring	31
Support.....	32
Billing and accounting.....	32
Looking at the Technical Interface.....	32
APIs and data transformations.....	33
Data and application architecture.....	33
Security in the cloud	34
Managing Cloud Resources	34
IT security.....	35
Performance management.....	35
Provisioning.....	36
Service management	37
Untangling Software Dependencies.....	37

Chapter 4: Developing Your Cloud Strategy 39

Seeing the Many Aspects of Your Cloud Strategy	40
Questioning Your Company's Strategy.....	41
Assessing Where You Are Today	42
How tangled is my computing environment?.....	42
What's my data center environment?	42
What data supports my strategy?	43
Assessing Your Expense Structure	44
Checking Up on Rules and Governances.....	44
Developing a Road Map	45

Part II: Understanding the Nature of the Cloud 47

Chapter 5: Seeing the Advantages of the Highly Scaled Data Center 49

Comparing Financial Damage: Traditional versus Cloud	50
Traditional data center	50
Cloud data center	51
Scaling the Cloud.....	52
Comparing Traditional and Cloud Data Center Costs	55

Examining labor costs and productivity.....	56
Wondering where you are	56

Table of Contents **XV**

Chapter 6: Exploring the Technical Foundation for Scaling Computer Systems 59

Server-ing Up Some Hardware	60
Tradition! versus clouds	60
Considering cloud hardware.....	61
Open-source dynamic	63
Economies of Scale.....	63
Benefitting enormously.....	64
Optimizing otherwise	64
Keeping the Bottom Line in	

Chapter 7: Checking

the Cloud's Workload Strategy 67

Managing Workloads in the Cloud	67
Thinking of workloads as well-planned services	68
Creating interfaces between containers	70
Discovering how XML fits in.....	70
Using container workloads: Case study	71
Balancing Risk and Practical Models	71
Testing Workloads in the Real World	73

Chapter 8: Managing Data 75

Declaring Data Types	75
Securing Data in the Cloud.....	76
Data location in the cloud.....	77
Data control in the cloud.....	78
Securing data for transport in the cloud	79
Looking at Data, Scalability, and Cloud Services.....	81
Large-scale data processing	81
Databases and data stores in the cloud.....	82
Data archiving	84
Sorting Out Metadata Matters	84
Talking to Your Cloud Vendor about Data.....	84

Chapter 9: Discovering Private and Hybrid Clouds 87

Pining for Privacy	88
Defining a private cloud.....	88
Comparing public, private, and hybrid.....	89
Examining the Economics of the Private	

Cloud.....	92	Assessing capital expenditures
.....	92	
Vendor private cloud offerings	93	
Offering Up Key Vendors	94	Services-led
.....	95	technology companies
Systems integrators companies.....	98	
Technology enabler companies	99	

XVI Cloud Computing For Dummies

Part III: Examining the Cloud Elements..... 105

Chapter 10: Seeing Infrastructure as a Service 107

Tracing IaaS to ISP	107
Renting (but not to own)	108
Following the ISP pattern.....	109
Exploring Amazon EC2: Case Study	109
EC2 Compute Units	110
Platforms and storage	110
EC2 pricing.....	112
EC2 customers.....	112
Checking Out Other IaaS Companies	113
Rackspace	113
GoGrid	114
Others.....	114
Examining IaaS-Enabling Technology	114
AppLogic	115
Eucalyptus	115
Trusting the Cloud.....	116
What Infrastructure as a Service Means to You	117

Chapter 11: Exploring Platform as a Service 119

Putting Platform as a Service on a Pedestal.....	120
Integrated lifecycle platforms	121
Anchored lifecycle platforms	122
Enabling technologies as a platform	122
Getting Inside the Integrated Lifecycle Platform.....	122
Google App Engine.....	123
Microsoft Azure.....	125
Getting Inside Anchored Lifecycle Platform as a Service.....	127
Salesforce.com's Force.com platform.....	127
Intuit	130
LongJump	132
Enabling Technologies as a Platform.....	133
Testing in the cloud.....	134
Service management for the cloud.....	134
Integration and configuration platforms.....	134
Social network, framework, and portal platforms.....	135

Chapter 12: Using Software as a Service 137

SalesForce.com's Approach to Evolving Software as a Service	138
Salesforce.com software environment	138
SalesForce.com ecosystem	140
Characterizing Software as a Service.....	140

Understanding the Economics and the Ecosystem.....	142
Pretending you're a customer.....	142
The value of the ecosystem.....	144

Table of Contents xvii

Examining Types of SaaS Platforms	145
Packaged Software as a Service	147
Collaboration as a Service	148
Enabling and management tools.....	149

Chapter 13: Understanding Massively Scaled Applications and Business Processes 153

Naming Names: Companies with Massively Scaled Applications	154
Listing the companies	154
Looking at Web-based business services	156
Delivering Business Processes from the Cloud.....	157
Business process examples.....	157
Business processes destined for the cloud.....	158
Hidden in the cloud	158
Business processes already flying high	158
Predicting the future	159

Chapter 14: Setting Some Standards 161

Understanding Best Practices and Standards	161
Best practicing makes perfect.....	162
Setting your sites on standards	162
Clouding the Standards and Best Practices Issue.....	163
Interoperability	164
Portability	164
Integration	164
Security	164
Standards Organizations and Groups.....	166
Cloud Security Alliance.....	166
Distributed Management Task Force (DMTF).....	167
National Institute of Standards and Technology (NIST).....	167
Open Cloud Consortium (OCC).....	168
Open Grid Forum (OGF).....	168
The Object Management Group (OMG)	169
Storage Networking Industry Association (SNIA).....	169
Cloud Computing Interoperability Forum (CCIF)	169
Vertical groups.....	170

Part IV: Managing the Cloud..... 171

Chapter 15: Managing and Securing Cloud Services 173

Putting Security on the Spot with Questions	174
Understanding Security Risks.....	175
Reducing Cloud	

Security Breaches	177	Implementing
Identity Management	179	Benefits of
identity management.....	179	
Aspects of identity management	180	

xviii Cloud Computing For Dummies

Playing Detective: Detection and Forensics.....	182
Activity logs.....	182
HIPS and NIPS.....	182
Data audit.....	184
Encrypting Data	184
Creating a Cloud Security Strategy	185

Chapter 16: Governing the Cloud 187

Looking at IT Governance	188
Deciding on a Governor	189
Imagining a scenario.....	190
Imagining another scenario.....	190
Knowing the Risks of Running in the Cloud.....	190
Understanding risk	191
Measuring and monitoring performance.....	193
Measurement methods	193
Making Governance Work	194
Establishing your governance body.....	194
Monitoring and measuring IT service performance.....	195
Cataloging control and compliance data.....	195

Chapter 17: Virtualization and the Cloud 197

Visualizing Virtualization.....	197
Characteristics	198
Using a hypervisor in virtualization	199
Abstracting hardware assets.....	201
Managing Virtualization.....	202
Foundational issues.....	202
Abstraction layer	203
Provisioning software.....	204
Virtualizing storage	205
Hardware provisioning.....	205
Security issues	206
Taking Virtualization into the Cloud.....	208

Chapter 18: Managing Desktops and Devices in the Cloud 209

Virtualizing the Desktop	209
Across industries.....	210
The client desktop	210
Putting Desktops in the Cloud	212
Further pros.....	213
Desktop as a Service (DaaS)	213
Managing Desktops in the Cloud.....	215
Watching four areas	215
Managing assets.....	216

Monitoring services	217
Change management.....	218
Security	218
Getting a Reality	

Check.....	219	Chapter 19:
------------	-----	--------------------

Service Oriented Architecture and the Cloud 221

Defining Service Oriented	
Architecture.....	221
cloud and SOA	222
Characterizing SOA.....	222
Loosening Up on Coupling	
.....	223
Making SOA	
Happen.....	224
Catching	
the Enterprise Service Bus.....	225
Telling your registry from your repository	225
Cataloging services.....	227
Understanding Services in the	
Cloud.....	228
Serving the Business with	
SOA and Cloud Computing	230

Chapter 20: Managing the Cloud Environment 231

Managing the	
Cloud.....	232
The	
service provider	232
Customers.....	234
Hybrid environments	236
Building Up Support	
Desks.....	237
Service	
desk goals.....	237
Varying support levels	238
Examining support services	238
Gaining	
Visibility.....	240
Ensuring adequate performance levels	
.....	241
Monitoring service availability	241
Tracking Service Level Agreements	

.....241 Part V: Planning for the

Cloud..... 243

Chapter 21: Banking on Cloud Economics 245

Seeing the Cloud's	
Allure.....	245
Filling the	
need for capacity.....	246
Getting the work done without capital investment.....	246
Selecting a SaaS for common applications	247

Selecting the massively scaled application.....	247
When it's not black and white.....	247
Creating an Economic Model of the Data Center	248
Listing application costs.....	248
Recovering costs.....	250

XX Cloud Computing For Dummies

Adjusting the Economic Model even Further	251
Private cloud and allocation costs	251
Service levels and compliance costs.....	252
Strategic considerations and costs	253
Summarizing an Economic Cost Model	253

Chapter 22: Starting Your Journey to the Cloud 255

Putting the Kibosh on Cloud Cultural Issues	255
Anticipating (but not with relish).....	256
Smoothing the transition	257
Measuring Twice: Assessing Risks	258
Playing risk with categories.....	258
Top company concerns	259
Picking the Right Targets for Success.....	260
Picking the low-hanging fruit.....	261
Approaching other areas	261
Planning for Leveraging the Cloud.....	262
Example 1.....	262
Example 2.....	263

Part VI: The Part of Tens..... 265

Chapter 23: Ten (Plus One) Swell Cloud Computing Resources . . . 267

Hurwitz & Associates	267
National Institute of Standards and Technology	268
CloudCamp.....	268
SaaS Showplace.....	268
TechTarget.....	268
The Cloud Standards Wiki.....	269
Finding OASIS.....	269
The Eclipse Foundation	269
The Cloud Security Alliance.....	269
Open Cloud Manifesto	270
Vendor Sites	270

Chapter 24: Ten Cloud Dos and Don'ts 271

Don't Be Reactive	271
Do Consider the Cloud a Financial Issue	271
Don't Go It Alone.....	272
Do Think about Your Architecture.....	272
Don't Neglect Governance.....	272

Don't Forget about Business Process	272
Do Make Security the Centerpiece of Your Strategy.....	273
Don't Apply the Cloud to Everything.....	273
Don't Forget about Service Management	273
Do Start with a Pilot Project.....	273

Glossary.....	275
----------------------	------------

Index.....	
------------	--

291

xxii Cloud Computing For Dummies

Introduction

Welcome to *Cloud Computing For Dummies*. You can't read a technology journal or blog — or even your local newspaper — without coming upon a reference to cloud computing. While there's been a lot of debate about what cloud computing is and where it's headed, no one has doubts that it is real.

In fact, we think that cloud computing, in all of its forms, is transforming the computing landscape. It will change the way we deploy technology and how we think about the economics of computing. We hope this book provides a perspective on cloud computing and starts your journey of exploration.

Cloud computing is more than a service sitting in some remote data center. It's a set of approaches that can help organizations quickly, effectively add and subtract resources in almost real time. Unlike other approaches, the cloud is as much about the business model as it is about technology. Companies clearly understand that technology is at the heart of how they operate their businesses. Business executives have long been frustrated with the complexities of getting their computing needs met quickly and cost effectively. In a sense, cloud computing has started to become mainstream because these business executives have forced the issue into the forefront.

Cloud computing isn't a quick fix. It requires a lot of thought: Which approach is most appropriate for your company? For example, companies have to decide if they want to use *public* (external) cloud services or if they want to have private clouds behind their firewalls. How should you architect your internal environment to support the cloud?

The cloud environment itself requires a strong foundation of best practices in software development, software architecture, and service management foundations. This strong foundation is especially important because most organizations combine public and private cloud services. You want to be informed before you start your search. We think this book will give you the context to make informed decisions.

2 Cloud Computing For Dummies

About This Book

Cloud computing is a big new area and requires that a lot of people get familiar with it in a fairly short period of time. That's why we wrote this book. Some people may want to get deeper into the technological details, while others may care only about the business implications.

We recommend that you read the first five chapters, regardless of how deeply you want to wander into the cloud. These chapters give you context about the cloud and what's behind the concept. If you want to begin understanding the economics and the available approaches to the cloud, you should read the later chapters.

You can read from cover to cover, but if you're not that kind of person, but we've tried to adhere to the *For Dummies* style of keeping chapters self-contained so you can go straight to the topics that interest you most. Wherever you start, we wish you well.

Foolish Assumptions

Try as we might to be all things to all people, when it came to writing this book, we had to pick who we thought would be most interested in *Cloud Computing For Dummies*. Here's who we think you are:

- ✓ **You're smart.** You're no dummy, yet the topic of service oriented architecture gives you an uneasy feeling; you can't quite get your head around it, and if you're pressed for a definition, you might try to change the subject.
- ✓ **You're a businessperson who wants little or nothing to do with technology,** but you live in the 21st century and find that you can't escape it. Everybody's saying, "It's all about moving to the cloud," so you think that you better find out what they're talking about.
- ✓ **You're an IT person who knows a heck of a lot about technology,** but who is new to this cloud stuff. Everybody says it's something different. Once and for all, you want the whole picture.

Whoever you are, welcome. We're here to help.

How This Book Is Organized

We divide our book into six parts for easy consumption. Feel free to skip about.

Introduction 3

Part I: Introducing Cloud Computing

In this part, we explain the fundamentals of cloud computing from a business and technology perspective. We also introduce you to the major concepts and components so you can hold your own in any meaningful cloud conversation.

Part II: Understanding the Nature of the Cloud

Part II is for more technically oriented people. In this section, we dive deeper into the actual foundational elements of the cloud.

Part III: Examining the Cloud Elements

What types of clouds are there and how do they work? This part delves into areas critical to your cloud plans.

Part IV: Managing the Cloud

The rubber meets the road right here. A cloud computing environment can't work if it isn't well managed. This section gives you plenty to think about in this critical area.

Part V: Planning for the Cloud

When you understand what the cloud is all about, you can start planning. You need to think about the financial implications of clouds as well as the steps that get you going.

Part VI: The Part of Tens

If you're new to the *For Dummies* treasure trove, you're no doubt unfamiliar with "The Part of Tens." In "The Part of Tens," Wiley editors torture *For Dummies* authors into creating useful bits of information easily accessible in lists containing ten (or more) elucidating elements. We started these chapters kicking and screaming but are ultimately very glad they're here. We think you'll be glad, too.

Icons Used in This Book

Pay attention. The bother you save may be your own.

You may be sorry if this little tidbit slips your mind.

We think this a particularly useful point to pay attention to.

Tidbits for the more technically inclined.

Where to Go from Here

We've created an overview of cloud computing and introduce you to all of its significant components. Many of these chapters could be expanded into full length books of their own. Cloud computing and the entire distributed technology landscape is a big focus for us at Hurwitz & Associates, and we invite you to visit our site and read our blogs and insights at www.hurwitz.com.

Part I

Introducing Cloud Computing

The 5th Wave

By Rich Tennant



"Oh, I'll get us in - I used to run tech support at an internet access company."

In this part . . .

The phrase *cloud computing* implies something, but

exactly what is it? In this part, we introduce the concept and provide a simple graphic that describes the layers of cloud computing. We also examine the value that the cloud can bring to your organization and look at some of the associated business issues.

Chapter 1

Grasping the Fundamentals

In This Chapter

- Doing your computing on the cloud
- Seeing what the cloud's made of
- Comparing the cloud to tradition
- Driving your business

In a dynamic economic environment, your company's survival may depend

on your ability to focus on core business and adapt quickly. Yesterday's profitable business model can't be counted on to translate into future growth and profits. As your business adapts to changing government and industry regulations, evaluates new business partnerships, and anticipates competitive threats, IT needs to help the business find new ways to respond.

At the same time, plans for change must often be made in the context of limited resources for finances, people, technology, and power. In this chapter, we introduce you to cloud computing — what it is and how it helps companies rethink how they deploy technology.

While there are a lot of technical considerations, keep in mind the fundamental truth: Cloud computing is a business and economic model. Is cloud computing a replacement for the traditional data center? The answer is complicated. In some cases, yes; in some cases, no.

Are we suggesting that the traditional data center goes away to be replaced with a cloud? Not necessarily. Sometimes the traditional data center is the best fit. However, for business agility and economic reasons, the cloud is becoming an increasingly important option for companies. We see cloud computing as the foundation for the industrialization of computing. Yes, it is that important.

8 Part I: Introducing Cloud Computing

Considering Perspectives

In this book, we look at cloud computing from three perspectives: the strategy from both the customer and the provider's point of view, business and economic considerations, and the technical underpinnings. We also examine how companies are using the cloud to control IT expenditures as they prepare to move to a service-centric world.

Many players make up the world of cloud computing:

- ✓ The **vendors** providing applications and enabling technology, infrastructure, hardware, and integration
- ✓ The **partners** of these vendors that are creating cloud services offerings and providing support services to customers
- ✓ The **business leaders** themselves who are either using or evaluating

various types of cloud computing offerings

This book addresses each of these audiences because they're all a fundamental part of this fabric of the future of computing.

Computing on the Cloud

What is cloud computing? Cloud computing is the next stage in evolution of the Internet. The *cloud* in cloud computing provides the means through which everything — from computing power to computing infrastructure, applications, business processes to personal collaboration — can be delivered to you as a service wherever and whenever you need.

Cloud computing is offered in different forms:

- ✓ Public clouds
- ✓ Private clouds
- ✓ Hybrid clouds, which combine both public and private

In general the cloud — similar to its namesake of the cumulus type — is fluid and can easily expand and contract. This *elasticity* means that users can request additional resources on demand and just as easily *deprovision* (or release) those resources when they're no longer needed. This elasticity is one of the main reasons individual, business, and IT users are moving to the cloud.

In the traditional data center it has always been possible to add and release resources. However, this process couldn't be done in an automated or self service manner.

Chapter 1: Grasping the Fundamentals 9

This evolution to cloud computing — already underway — can completely change the way companies use technology to service customers, partners, and suppliers. Some businesses already have IT resources almost entirely in the cloud. They feel that the cloud model provides a more efficient, cost effective IT service delivery.

This doesn't mean that all applications, services, and processes will necessarily be moved to the cloud. Many businesses are much more cautious and are taking a hard look at their most strategic business processes and intellectual property to determine which computing assets need to remain under internal company control and which computing assets could be moved to the cloud.

Defining the Cloud

The *cloud* itself is a set of hardware, networks, storage, services, and interfaces that enable the delivery of computing as a service. *Cloud services* include the delivery of software, infrastructure, and storage over the Internet (either as separate components or a complete platform) based on user demand.

The world of the cloud has lots of participants:

- ✓ The **end user** doesn't really have to know anything about the underlying technology. In small businesses, for example, the cloud provider becomes the de facto data center. In larger organizations, the IT organization oversees the inner workings of both internal resources and external cloud resources.
- ✓ **Business management** needs to take responsibility for overall governance of data or services living in a cloud. Cloud service providers must provide a predictable and guaranteed service level and security to all their constituents.
- ✓ The **cloud service provider** is responsible for IT assets and maintenance.

Therefore, we have written this book to include the concerns of all the players in the evolving cloud ecosystem.

Cloud services must enable *multi-tenancy* — different companies sharing the same underlying resources. This topic is discussed further in Chapter 12.

Companies are finding some important new value in cloud services. The cloud can eliminate many of the complex constraints from the traditional computing environment, including space, time, power, and cost.

Cloud services like social networks (such as Facebook or LinkedIn) and collaboration tools (like video conferencing, document management, and webinars) are changing the way people in businesses access, deliver, and

10 Part I: Introducing Cloud Computing

understand information. Cloud computing infrastructures make it easier for companies to treat their computing systems as a pool of resources rather than a set of independent environments that each has to be managed.

Overall, the cloud embodies the following four basic characteristics:

- ✓ Elasticity and the ability to scale up and down
- ✓ Self-service provisioning and automatic deprovisioning
- ✓ Application programming interfaces (APIs)
- ✓ Billing and metering of service usage in a pay-as-you-go model

Each of these characteristics is described in more detail in the following sections.

Elasticity and scalability

The service provider can't anticipate how customers will use the service. One customer might use the service three times a year during peak selling seasons, whereas another might use it as a primary development platform for all of its applications.

Therefore, the service needs to be available all the time (7 days a week, 24 hours a day) and it has to be designed to scale upward for high periods of

demand and downward for lighter ones. *Scalability* also means that an application can scale when additional users are added and when the application requirements change.

This ability to scale is achieved by providing *elasticity*. Think about the rubber band and its properties. If you're holding together a dozen pens with a rubber band, you probably have to fold it in half. However, if you're trying to keep 100 pens together, you will have to stretch that rubber band. Why can a single rubber band accomplish both tasks? Simply, it is elastic and so is the cloud.

In Chapter 2, we give you some concrete examples of how providers are using this characteristic.

Self-service provisioning

Customers can easily get cloud services without going through a lengthy process. The customer simply requests an amount of computing, storage, software, process, or other resources from the service provider. Chapter 7 explains this process in detail.

Chapter 1: Grasping the Fundamentals 11

Contrast this on-demand response with the process at a typical data center. When a department is about to implement a new application, it has to submit a request to the data center for additional computing hardware, software, services, or process resources. The data center gets similar requests from departments across the company and must sort through all requests and evaluate the availability of existing resources versus the need to purchase new hardware. After new hardware is purchased, the data center staff has to configure the data center for the new application. These internal procurement processes can take a long time, depending on company policies.

Of course, nothing is as simple as it might appear. While the on-demand provisioning capabilities of cloud services eliminates many time delays, an organization still needs to do its homework. These services aren't free; needs and requirements must be determined before capability is automatically provisioned.

Application programming interfaces (APIs)

Cloud services need to have standardized APIs. These interfaces provide the instructions on how two application or data sources can communicate with each other.

A standardized interface lets the customer more easily link a cloud service, such as a customer relationship management system with a financial accounts management system, without having to resort to custom programming. For more information on standards see Chapter 14.

Billing and metering of services

Yes, there is no free lunch. A cloud environment needs a built-in service that bills customers. And, of course, to calculate that bill, usage has to be *metered* (tracked). Even free cloud services (such as Google's Gmail or Zoho's Internet-based office applications) are metered.

In addition to these characteristics, cloud computing must have two overarching requirements to be effective:

- ✓ A comprehensive approach to service management
- ✓ A well-defined process for security management

12 Part I: Introducing Cloud Computing

Performance monitoring and measuring

A cloud service provider must include a service management environment. A *service management environment* is an integrated approach for managing your physical environments and IT systems. This environment must be able to maintain the required service level for that organization.

In other words, service management has to monitor and optimize the service or sets of services. Service management has to consider key issues, such as performance of the overall system, including security and performance. For example, an organization using an internal or external email cloud service would require 99.999 percent uptime with maximum security. The organization would expect the cloud provider to prove that it has met its obligations.

Many cloud service providers give customers a dashboard — a visualization of key service metrics — so they can monitor the level of service they're getting from their provider. Also, many customers use their own monitoring tools to determine whether their service level requirements are being met.

Security

Many customers must take a leap of faith to trust that the cloud service is safe. Turning over critical data or application infrastructure to a cloud-based service provider requires making sure that the information can't be accidentally accessed by another company (or maliciously accessed by a hacker).

Many companies have compliance requirements for securing both internal and external information. Without the right level of security, you might not be able to use a provider's offerings. For more details on security, see Chapter 15.

Comparing Cloud Providers with

Traditional IT Service Providers

Traditional IT service providers operate the hardware, software, networks, and storage for its clients. While the customer pays the licensing fees for the software, the IT service provider manages the overall environment. The service provider operates the infrastructure in its own facilities. With the traditional IT service provider, the customer signs a long-term contract that specifies mutually agreed-upon service levels. These IT providers typically customize an environment to meet the needs of one customer.

Chapter 1: Grasping the Fundamentals 13

In the cloud model, the service provider might still operate the infrastructure in its own facilities (except in the case of a private cloud, which we discuss in Chapter 9). However, the infrastructure might be *virtualized* across the globe, meaning that you may not know where your computing resources, applications, or even data actually reside. (We talk more about virtualization in Chapter 17.) Additionally, these service providers are designing their infrastructure for scale, meaning that there isn't necessarily a lot of customization going on. (We talk more about the scale issue in Chapter 13.)

Addressing Problems

There is an inherent conflict between what the business requires and what data center management can reasonably provide. Business management wants optimal performance, flawless implementation, and 100 percent uptime. The business leadership wants new capability to be available immediately, frequent changes to applications, and more accessibility to quality data in real time — but their organizations have limited budgets.

Getting on board with cloud computing

Although opinions differ about how quickly technology will migrate to the cloud, without doubt the interest level is high. Lots of business folks are asking questions about the cloud approach when they hear about the data center efficiencies achieved by companies like Amazon (www.amazon.com) and Google (www.google.com).

For example, a smart CEO was under a lot of pressure to improve profitability by cutting capital expenditures. One day he read an article about the economic advantages of cloud computing in a business journal and began to wonder, "Hey, if Amazon can offer computing on demand, why can't our own IT department

act

like that?" The CEO paid a visit to the CIO and asked that very question. The CIO wasn't quite sure how to answer his boss. His only reply was that things are more complicated than that. The CIO pointed out issues related to data security and privacy. In addition, there are applications running in the data center that are one-of-a-kind and not easily handled. At the same time, he recognized that the department needed to provide better service to internal customers. The CIO did agree that there were other areas of IT that might be appropriate for the cloud model. For example, areas such as testing, software development, storage, and email were good candidates for cloud computing.

Over time, it became easier for IT to add hardware to the data center rather than to focus on making the data center itself more effective. And this plan worked. By pouring more resources into the data center, IT ensured that critical applications wouldn't run out of resources. At the same time, these companies built or bought software to meet business needs. The applications that were built internally were often large and complex. They had been modified repeatedly to satisfy changes without regard to their underlying architecture.

Between managing a vast array of expanding hardware resources combined with managing huge and unwieldy business software, IT management found itself under extraordinary pressure to become much more effective and efficient.

This tug of war between the needs of the business and the data center constraints has caused friction over the past few decades. Clearly, need and money must be balanced. To meet these challenges, there have been significant technology advancements including virtualization (see Chapter 17), service-oriented architecture (see Chapter 19), and service management (see Chapter 20). Each of these areas is intended to provide more modularity, flexibility, and better performance for IT.

While these technology enablers have helped companies to become more efficient and cost effective, it isn't enough. Companies are still plagued with massive inefficiencies. The promise of the cloud is to enable companies to improve their ability to leverage what they've bought and make use of external resources designed to be used on demand.

We don't want to give you the idea that everything will be perfect when you get yourself a cloud. The world, unfortunately, is more complicated than that. For example, complex, brittle applications won't all be successful if they are just thrown up on the cloud. Virtualization adds performance implications. And many of these applications lack an architecture to achieve scale. A database-bound application will remain database bound, regardless of the additional compute resources beneath it.

Discovering the Business Drivers for Consuming Cloud Services

In the beginning of this chapter, we name reasons companies are thinking about cloud services and some of the pressures coming from management. Clearly, business management is under a lot of pressure to reduce costs while providing a sophisticated level of service to internal and external customers. In this section, we talk about the benefits of cloud services.

One of the most immediate benefits of cloud-based infrastructure services is the ability to add new infrastructure capacity quickly and at lower costs. Therefore, cloud services allow the business to gain IT resources in a self-service manner, thus saving time and money. By being able to move more quickly, the business can adapt to changes in the market without complex procurement processes.

A typical cloud service provider has *economies of scale* (cost advantages resulting in the ability to spread fixed costs over more customers) that the typical corporation lacks. As mentioned earlier, the cloud's self-service capability means it's easier for IT to add more *compute cycles* (more CPU resources added on an incremental basis) or storage to meet an immediate or intermittent needs.

With the advent of the cloud, an organization can try out a new application or develop a new application without first investing in hardware, software, and networking.

Reducing capital expenditures

You might want to add a new business application, but lack the money. You might need to increase the amount of storage for various departments. Cloud service providers offer this type of capability at a prorated basis. A cloud service vendor might rent storage on a per-gigabyte basis.

Companies are often challenged to increase the functionality of IT while minimizing capital expenditures. By purchasing just the right amount of IT resources on demand the organization can avoid purchasing unnecessary equipment. There are always trade-offs in any business situation.

A company may significantly reduce expenses by moving to the cloud and then may find that its operating expenses increase more than predicted. In other situations, the company may already have purchased significant IT resources and it may be more economically efficient to use them to create a private cloud. Some companies actually view IT as their primary business and therefore will view IT as a revenue source. These companies will want to invest in their own resources to protect their business value.

16 Part I: Introducing Cloud Computing

Chapter 2

Discovering the Value of the Cloud for Business

In This Chapter

- Introducing a model of the cloud

- ▶ Getting familiar with *as a service*
- ▶ Measuring the cloud value to your business

As soon as you start reading about cloud computing, you run into the words *as a service* an awful lot. Examples include *Infrastructure as a Service*, *hardware as a Service*, *social networks as a service*, *applications as a service*, *desktops as a service*, and so on.

The term *service* is a task that has been packaged so it can be automated and delivered to customers in a consistent and repeatable manner. These services may be delivered by a cloud service vendor or through your own internal data center.

Modeling Services

We include the various types of cloud services into three distinct models, illustrated as different layers in Figure 2-1. The reality is that there is a blend in between the types of service delivery models that are available from cloud vendors. For example, a Software as a Service vendor might decide to offer separate infrastructure services to customers. The purpose of grouping these services into three models is to aid in understanding what lies beneath a cloud service. All these service delivery models require management and administration (including security), as depicted by the outer ring in Figure 2-1.

18 Part I: Introducing Cloud Computing

The three cloud service delivery models are Infrastructure as a Service, Platform as a Service, and Software as a Service, and the purpose of each model is as follows:

- ✓ The Infrastructure as a Service layer offers storage and compute resources that developers and IT organizations use to deliver custom business solutions.
- ✓ The Platform as a Service layer offers development environments that IT organizations can use to create cloud-ready business applications.
- ✓ The Software as a Service layer offers purpose-built business applications.

In this chapter we provide an introduction to each model. In addition, because an understanding of each model is critical to developing an understanding of cloud computing, each model is covered in separate chapters in Part II.

The customer accesses those services with defined interfaces. These interfaces are, in fact, all that the user ever comes in contact with. The customer never sees the infrastructure that provides a movie on demand, for example — they only see the screen that enables the user to select and purchase the movie. Likewise, in cloud computing the underlying infrastructure that provides the service may be very sophisticated indeed. However, the user doesn't necessarily need to

understand this infrastructure to use it.

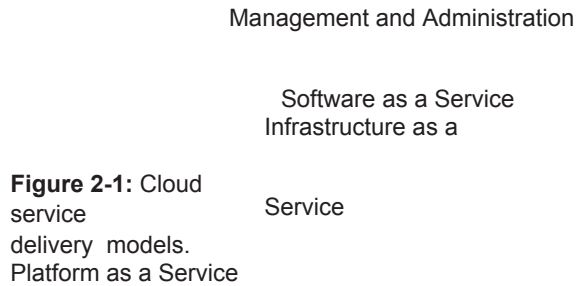


Figure 2-1: Cloud service delivery models.

Understanding Infrastructure as a Service

Infrastructure as a Service (IaaS) is the delivery of computer hardware (servers, networking technology, storage, and data center space) as a service. It may also include the delivery of operating systems and virtualization technology to manage the resources.

Chapter 2: Discovering the Value of the Cloud for Business 19

The IaaS customer rents computing resources instead of buying and installing them in their own data center. The service is typically paid for on a usage basis. The service may include *dynamic scaling* so that if the customer winds up needing more resources than expected, he can get them immediately (probably up to a given limit).

Dynamic scaling as applied to infrastructure means that the infrastructure can be automatically scaled up or down, based on the requirements of the application.

Additionally, the arrangement involves an agreed-upon *service level*. The service level states what the provider has agreed to deliver in terms of availability and response to demand. It might, for example, specify that the resources will be available 99.999 percent of the time and that more resources will be provided dynamically if greater than 80 percent of any given resource is being used.

Currently, the most high-profile IaaS operation is Amazon's Elastic Compute Cloud (Amazon EC2). It provides a Web interface that allows customers to access virtual machines. EC2 offers scalability under the user's control with the user paying for resources by the hour. The use of the term *elastic* in the naming of Amazon's EC2 is significant. The elasticity refers to the ability that EC2 users have to easily increase or decrease the infrastructure resources assigned to meet their needs. The user needs to initiate a request, so this service provided isn't dynamically scalable. Users of EC2 can request the use of any operating system as long as the developer does all the work. Amazon itself supports a more limited number of operating systems (Linux, Solaris, and Windows). For an up-to-the-minute description of this service, go to <http://>

Service delivery models defined

You have probably noticed a multitude of companies providing all kinds of cloud services, using their own resources. Services you purchase from these cloud service providers are offered to you the same way your TV cable provider offers services. Your cable contract

provides you with access to watch a specific set of television channels. In addition to receiving your standard channels, you may have a self-service option where you can purchase a movie to watch on demand.

20 Part I: Introducing Cloud Computing

Companies with research-intensive projects are a natural fit for IaaS. Cloud based computing services allow scientific and medical researchers to perform testing and analysis at levels that aren't possible without additional access to computing infrastructure.

Other organizations with similar needs for additional computing resources may boost their own data centers by renting the computer hardware — appropriate allocations of servers, networking technology, storage, and data center space — as a service. Instead of laying out the capital expenditure for the maximum amount of resources to cover their highest level of demand, they purchase computing power when they need it.

Exploring Platform as a Service

With *Platform as a Service (PaaS)*, the provider delivers more than infrastructure. It delivers what you might call a *solution stack* — an integrated set of software that provides everything a developer needs to build an application — for both software development and runtime.

PaaS can be viewed as an evolution of Web hosting. In recent years, Web hosting companies have provided fairly complete software stacks for developing Web sites. PaaS takes this idea a step farther by providing *lifecycle management* — capabilities to manage all software development stages from planning and design, to building and deployment, to testing and maintenance. The primary benefit of PaaS is having software development and deployment capability based entirely in the cloud — hence, no management or maintenance efforts are required for the infrastructure. Every aspect of software development, from the design stage onward (including source-code management, testing, and deployment) lives in the cloud.

PaaS is inherently multi-tenant and naturally supports the whole set of Web services standards and is usually delivered with dynamic scaling. In reference to Platform as a Service, *dynamic scaling* means that the software can be automatically scaled up or down. Platform as a Service typically addresses the need to scale as well as the need to separate concerns of access and data security for its customers.

Although this approach has many benefits for customers, it also has some disadvantages. The major drawback of Platform as a Service is that it may lock you in to the use of a particular development environment and stack of software components. Platform as a Service offerings usually have some proprietary elements (perhaps the development tools or even component libraries). Consequently, you may be wedded to the vendor's platform and unable to move your applications elsewhere without rewriting them to some degree. If you suddenly become dissatisfied with your Platform as a Service provider, you may face very high expenses when you suddenly need to rewrite the applications to satisfy the requirements of another PaaS vendor.

Chapter 2: Discovering the Value of the Cloud for Business 21

The fear of vendor lock-in has led to a new variety of Platform as a Service emerging: Open Platform as a Service. This would offer the same approach as Platform as a Service, except that there is no constraint on choice of development software. It avoids the possibility of lock-in.

Some examples of Platform as a Service include the Google App Engine, AppJet, Eteios, Qrimp, and Force.com, which is the official development environment for Salesforce.com. See the “Salesforce.com and automation application” sidebar elsewhere in this chapter for more on this pioneering example of Platform as a Service.

Seeing Software as a Service

One of the first implementations of cloud services was *Software as a Service (SaaS)* — business applications that are hosted by the provider and delivered as a service.

SaaS has its roots in an early kind of hosting operation carried out by *Application Service Providers (ASPs)*. The ASP business grew up soon after the Internet began to mushroom, with some companies offering to securely, privately host applications. Hosting of supply chain applications and *customer relationship management (CRM)* applications was particularly prominent, although some ASPs simply specialized in running email. Prior to the advent of this type of service, companies often spent huge amounts of money implementing and customizing these applications to satisfy internal business requirements. Many of these products weren't only difficult to implement but hard to learn and use. However, the most successful vendors were those who recognized that an application delivered as a service with a monthly fee based on the number of users had to be easy to use and easy to stay with.

CRM is one of the most common categories of Software as a Service; the most prominent vendor in this category is Salesforce.com, described in this chapter's sidebar. For a more extensive look at some of the other examples of Software as a Service, please refer to Chapter 12.

Buying Software as a Service offers a number of obvious advantages: While you can find a lot more information about these benefits in Chapter 12, the following provides some insight into why this approach to software delivery has gained so much traction with vendors and customers. The price of the

software is on a *per-use* basis and involves no upfront costs from the service provider. (Of course, the reality is that your company may have some upfront work to do to get your data loaded into the Software as a Service application database and you may have to deal with ongoing data integration between your internal and cloud data stores.) Businesses get the immediate benefit of reducing capital expenditures. In addition, a business gains the flexibility to test new software on a rental basis and then can continue to use and adopt the software, if it proves suitable.

22 Part I: Introducing Cloud Computing

Salesforce.com and automation application

Salesforce.com built and delivered a *sales force automation application* (which automates sales functions such as tracking sales leads and prospects and forecasting sales) that was suitable for the typical salesperson and built a business around making that application available over the Internet through a browser.

The company then expanded by encouraging the growth of a software ecosystem around its extended set of customer relationship management (CRM) applications, prompting other companies to integrate their business applications with those of Salesforce.com (or build components to add to Salesforce.com). It began, for example, by allowing customers to change tabs and create their own database objects. Next, the company added what it called the AppExchange, which added published application programming interfaces (APIs) so that third-party software providers could integrate their applications into the Salesforce.com platform.

Most AppExchange applications are more like utilities than full-fledged packaged apps. Many of the packages sold through the AppExchange are for tracking. For example, one tracks information about commercial and residential prop

erties; another optimizes the sales process for media/advertising companies; still another package analyzes sales data.

Salesforce.com took its offerings a step further by offering its own language called Apex. Apex is used only within the Salesforce.com platform and lets users build business applications and manage data and processes. A developer can use Apex to change the way the application looks. It is, in essence, the interface as a service.

With the advent of cloud computing, Salesforce.com has packaged these offerings into what it calls Force.com, which provides a set of common services its partners and customers can use to integrate into their own applications. Salesforce.com has thus started to also become a Platform as a Service vendor. Among the hundreds of applications that run on Force.com, it now offers a variety of HR software, and financial, supply chain, inventory, and risk management components. Just as Amazon is currently the trailblazer among the Infrastructure as a Service vendors, Salesforce.com is the trailblazer among the Software as a Service vendors. However, *many* vendors are now providing Applications as a Service. It has become a popular option for selling software.

Software as a Service modes

As a holdover from the traditional ASP model, Software as a Service comes in two distinct modes:

- ✓ **Simple multi-tenancy:** Each customer has its own resources that are segregated from those of other customers. It amounts to a relatively inefficient form of multi-tenancy.
- ✓ **Fine grain multi-tenancy:** This offers the same level of segregation but from a software engineering perspective, it's far more efficient. All resources are shared, but customer data and access capabilities are segregated within the application. This offers much superior economies of scale.

Chapter 2: Discovering the Value of the Cloud for Business 23

Initially, Software as a Service offerings were not simply implemented over the Internet. For the sake of security and reliability, these offerings would normally involve the use of *virtual private networks (VPNs)*. A VPN essentially makes the public network your own private network (by using some form of encryption) instead of having to purchase dedicated connectivity. This enables you to securely transmit data over a public network like the Internet.

Massively scaled Software as a Service

All as-a-service businesses are based on the service provider offering the service at a much lower cost than you providing it for yourself. If the price difference is large enough, assuming no other complications, it's a win-win — the provider grows a thriving business and the customers pay less to run their applications.

But some applications can be run *really* inexpensively in the cloud. When you have millions of users doing exactly the same thing — and we mean *exactly* the same thing (not similar things) — you can keep the cost per user very, very low. Enter *massively scaled Software as a Service*. One example is Yahoo Mail. Yahoo is the largest email provider, with approximately 260 million users.

This is possible because the provider can optimize all data center components including the hardware, communications, and software to support just one or two types of workloads.

Environments such as Facebook, eBay, Skype, Google Apps, and others are all designed for massive scaling. You may not think of many of these Web sites as being software applications at all. Nevertheless, all are used directly by businesses, for business purposes. For example, some companies use the social networking site Facebook as a free intranet for its employees. Online auctioneer eBay is the basis of more than 500,000 small businesses, Skype (free online calls and video) is used by small businesses the world over, and Google Apps (messaging and collaboration tools) has over a million different businesses enrolled. For more about this topic, take a look at Chapter 13.

Economies of scale

The companies that provide massively scaled Software as a Service achieve

dramatic economies of scale — cost efficiencies gained from reducing per-unit costs when more of the same item is produced or more of the same work loads are processed.

24 Part I: Introducing Cloud Computing

It's worth listing all the reasons why:

- ✓ The standardized workloads can be executed on a highly integrated, massively replicable infrastructure stack. They don't have to support a wide array of workloads and a heterogeneous stack of hardware, middle ware, OS, and so on.
- ✓ The computer hardware and network is highly streamlined and can be bought in bulk and configured to allow expansion. Often these companies require that hardware be engineered for their unique scaling requirements.
- ✓ All software can be stripped down so that only what is necessary is loaded.
- ✓ The service/software itself is written from scratch in a cloud-optimized way, tailored for efficiency at an instruction level.
- ✓ The provider may not offer or guarantee a specific service level.
- ✓ There is no need for virtualization technology to build virtual machines. The software can be engineered to the bare metal.
- ✓ The profile of the workload is measurable and predictable simply by numbers of users.

Management and Administration

If you refer to Figure 2-1, you will notice that the three layers are surrounded with an area called *Management and Administration*. This is where life in the cloud can get very complicated. It's simple enough to describe how to use some kind of cloud computing service, but you also have to integrate it into the IT operations of the organization, and that isn't necessarily a simple thing to do.

For example, because a cloud requires a self-service capability, it must be designed to manage not just provisioning customer requests but also issues such as workload management, security, metering, monitoring, and billing services. We provide much more detail on this topic in Chapters 21 and 22.

Many managers understand that for cloud services to be safe and effective, they must measure and monitor performance.

In fact, performance monitoring will become increasingly important as companies rely more on third-party services. And, from all indications, a typical company may use more than one cloud services provider. For example, a

Chapter 2: Discovering the Value of the Cloud for Business 25

company may use one cloud provider for a platform such as collaboration and a completely different provider for compute services. They may use

another provider for storage.

- ✓ How well does each cloud service perform?
- ✓ How are they performing together to support the business?
- ✓ Are the cloud services vendors adhering to governance rules that the company is required to follow?

Refer to Chapter 17 for more information on governance in the cloud.

Don't take a supplier's word that everything is working well. Although your company can save money in the data centers and on software licenses, you need to *spend* money and resources on service management to protect your business assets.

26 Part I: Introducing Cloud Computing

Chapter 3

Getting Inside the Cloud

In This Chapter

- Meeting organizational challenges
- Taking on administrative challenges
- Examining the technical interface
- Getting a handle on cloud resources
- Creating manageable services

At first glance, you might think that the cloud is a totally self-service environment. The reality is more complicated than that. The cloud, like every other computing platform, has to be managed. In this chapter, we discuss the overall cloud environment and the issues you need to consider, from organizational and administrative challenges to managing cloud resources.

Feeling Sensational about Organization

Cloud services impact your organization in subtle ways. The cloud impacts the whole company, not just the IT department:

- ✓ How do cloud services fit into your overall corporate and IT strategy? How will you manage cloud service providers along with your internal services? How will you make sure that your customers are well supported by services that are moving to a cloud?

- ✓ Does the cloud support your corporate and IT governance requirements?
- ✓ What are the important issues of emerging corporate and governmental standards, business process management, and the overall issues of managing costs?

28 Part I: Introducing Cloud Computing

Deciding on a strategy

Like any other technology strategy, a cloud strategy is considered in relationship to the following:

- ✓ Your IT organization's overall strategy
- ✓ Your company's overall strategy

You must make a complex evaluation of costs, benefits, business cultural issues, risks, and corporate and government standards before developing a *comprehensive* cloud strategy. Although very few organizations have tested cloud services in these heavy usage situations, a well-planned cloud service strategy has the potential to significantly reduce costs. Chapter 4 talks you through that strategizing.

Over time, however, as more well-tested commercial cloud services become available, companies will increasingly be able to rely on these services not just for IT cost savings, but also for delivering new value to the organization. The trend toward well-managed cloud services is especially important because of the increased automation across the organization. This may include the software embedded in everything from manufacturing systems to radio frequency identification tags that track inventory.

Cloud services can help organizations in steps. With *utility computing*, any customer can plug in an application or component because all the interfaces have been standardized between implementations. For companies to successfully use the cloud, management must decide what types of services they will begin deploying from the cloud.

One organization may decide that a Software as a Service approach is best, whereas another wants incremental capacity on demand. Before planning a usage strategy, consider what cloud services might be right for you. Most organizations adopt a hybrid strategy, combining internal managed services with cloud-based services. Chapter 9 details hybrid clouds.

Coping with governance issues

Four distinct cloud categories exist (and they're discussed at length in Chapter 2). Each approach presents different governance challenges:

- ✓ Infrastructure as a Service
- ✓ Platform as a Service
- ✓ Software as a Service
- ✓ Business Process as a Service

To make matters more complicated, these approaches have no clean dividing line. Emerging vendors often combine approaches into their offerings. In addition, in most instances, a hybrid situation develops where on-premise applications are used in collaboration with traditionally hosted services and cloud services.

Governing internally provided services and the externally provided cloud based services introduces new challenges for a company's strategy:

- ✓ How do you manage the overall lifecycle of your IT resources, including software licensing, cost allocation, and charge backs?
- ✓ How do you protect the integrity of your information resources? How do you ensure that you're complying with data privacy rules and regulations?
- ✓ How do you make sure that all your service providers can prove and document that they're meeting governmental and corporate requirements?

IT governance issues are complicated by new suppliers and new capabilities. With governance, your company needs to prove that it's complying with rules set by both governmental agencies and the corporation. Ideally, service providers of all types will deliver the same levels of control that you would have with your own resources. However, when you don't control how that new supplier operates, governance gets more complicated. Cloud computing requires a higher level of oversight to ensure that governance standards are met.

Monitoring business processes

Most cloud services impact the way business processes are implemented within an organization. For example, your organization may be using a cloud based service to check credit worthiness for potential customers. Therefore, you have to make sure that these services are linked back to your internal systems so things don't fall through the cracks.

Your business should standardize a way to monitor business processes that live entirely or partially in a cloud environment. An organization's important computer-dependent business processes need to be constantly monitored by software. Linking internal and external processes together in a seamless way is the best way to ensure customer satisfaction.

Many organizations already use third-party business process providers for things such as payment services. The importance of third-party providers continues to expand as more services are made available in the cloud — these services will be linked with a variety of internal and external providers. Software components of such business processes may migrate into the cloud, as long as this migration doesn't impede their monitoring. For that reason, you need to examine all cloud propositions to see if they impact business process monitoring.

Managing IT costs

All IT departments monitor costs, but few monitor them in terms of *asset performance* — the requirement to optimize the return on investments for both hardware and software. This is likely to change with the onset of cloud services. Unlike traditional licensing models, cloud propositions are based on rental arrangements.

You must compare two cost models:

- ✓ **Operating expenses** (paying per month, per user for each service)
- ✓ **Capital investments** (paying a purchase fee plus yearly maintenance for software that resides within your organization)

Evaluating the differences between the two cost models is a complex procedure for many companies. In some situations, the new cost models shift some responsibility away from IT to the business unit. For example, if a company's business unit hires 20 new employees and email is managed in the cloud, the business unit needs to budget for 20 more users. IT doesn't have to ensure that server capacity and IT staff are sufficient to support the additional users; that's now the responsibility of the cloud services provider. However, IT departments need to carefully monitor the effectiveness of the cloud environment to support the enterprise.

Administering Cloud Services

A company has to ask itself many questions:

- ✓ Are the cloud services doing what we want them to do?
- ✓ How do we know if the performance is at the right level?
- ✓ How can we judge whether the data that was deleted is really gone?

Solving these problems isn't easy. Investigating the reliability and viability of a cloud provider is one of the most complex areas faced when managing the cloud. The advent of cloud computing will be accompanied by disappointed customers and lawsuits for sure — some as a consequence of unrealistic expectations and some as a consequence of poor service.

It's particularly important for IT departments to enable administration systems that let them monitor every dimension of the service they're getting.

Chapter 3: Getting Inside the Cloud 31

In theory, the cloud service provider can build and provide a very stable service that is less expensive than a customer can implement internally. However, there can be a serious gap between the actual service and the promises made in the provider's sales literature.

You have to do your homework when evaluating the providers. Evaluate their experience in the market, the type of partnerships they've established, and their

reputation in the market. You can also talk to other customers that have used their services.

Here are some of the issues to consider:

- ✓ What vendors are available to solve your problem?
- ✓ How effective are the providers in managing their own environment?
- ✓ Do they provide repeatable services?
- ✓ How do these vendors handle an outage?
- ✓ What's their experience in dealing with customer issues?

In addition to finding a good partner, it's always a good idea to have more than one provider as an alternative.

Service level agreements and monitoring

Every company that buys any service from a cloud service provider must either accept a standard service level agreement (SLA) from the provider or negotiate such an agreement. A *service level agreement* is a contract that stipulates the type of service you need from providers and what type of penalties would result from an unexpected business interruption.

No organization should commit mission-critical systems to the cloud without negotiating an SLA that includes significant penalties for not delivering the promised service level. Management needs to know what service level is appropriate under changing business conditions. Management can't assume that the service provider will provide all the monitoring. Rather, the administrators must have their *own* ability to monitor service to satisfy the company's goals for performance.

32 Part I: Introducing Cloud Computing

Support

Support problems don't disappear when applications or infrastructures move to the cloud. You have to make sure that support targets are agreed on in advance with a cloud services provider. Therefore, your company must align its internal support team that deals with internal customers with the cloud provider.

What processes are in place to resolve problems when they arise? Just consider the situation where some important application has a performance problem. Especially in a hybrid environment, it's not always easy to tell if a problem resides within the cloud or outside of it. Such situations need to be prevented or at least dealt with very efficiently.

Billing and accounting

One cloud benefit is that, as a customer you can acquire just as much capability as needed. For this to work, billing and account management must be

automated. Customers, therefore, need to be able to monitor what they're using and how much it costs.

Potential problems arise if service level penalties aren't clear and if the provider adds too many incidental charges. Customers can run up unexpected bills if they can't accurately track usage.

Looking at the Technical Interface

Because the cloud service market is so new, few applications have been built from the ground up for this new environment. So far, no corporate applications were built with this model in mind.

Organizations that already have well-designed interfaces between application and infrastructure components may find it easier to transition to the cloud. Companies that have moved to a *service-oriented architecture (SOA)* are well positioned to make the move. Chapter 19 talks about SOA in detail.

With SOA, organizations build modular business services that include standardized interfaces. This modular approach is needed when approaching the highly distributed cloud environment. SOA is a good start; however, a lot of standardized interfaces will need to be developed for cloud service platforms in the coming years. (For more on the topic of service-oriented architecture, see *Service Oriented Architecture For Dummies*, Second Edition, by Judith Hurwitz, Robin Bloor, Marcia Kaufman, and Fern Halper.)

Chapter 3: Getting Inside the Cloud 33

APIs and data transformations

A cloud's *Application Programming Interface (API)* is the software interface that lets your company's infrastructure or applications plug in to the cloud. This is perhaps the most important place for standardization.

Many vendors in the cloud space would like to claim overall leadership and control over the interfaces. Therefore, many different vendors are developing their own interfaces. This, in turn, means that customers are likely to be forced to support multiple APIs. Managing multiple APIs means that when applications are changed, there's more programming involved; and there's more potential for errors when too many APIs are supported.

Even if vendors agree to a set of API standards, there will be data transformation issues (as data moves from one physical machine to another). These data transformations are the same as those required in projects such as building a data warehouse, with just one minor difference: The built-in software platform and Software as a Service environments must follow the data standards of the particular cloud service provider. For an organization to easily build connections between its internal data center and the cloud, it must use standardized APIs and data transformation capabilities.

Data and application architecture

New internally created services that support the changing business's changing demands must operate with cloud ecosystems. These services may need to migrate to and from the cloud. For example, a company might initiate a partnership that requires development and deployment in the cloud. This means that it will have to build an architecture that's modular enough to allow services to move between various cloud platforms.

The consistency and flexibility of an SOA approach makes it a good fit for the cloud. In an SOA environment, software components are put into services or containers. These containers hold software that executes a specific task. After software exists within a container, it can be ported from one environment to another, which makes it easier to port into and out of the cloud.

To be effective in a cloud environment, data also has to be packaged and managed. This isn't simple, but it's instrumental in making the cloud an effective business platform. The IT organization needs to manage data independently of the underlying packaged application, transactional system, or data environment such as a warehouse. Important data needs to easily move between internal data centers and external cloud-based environments. Your organization needs to start with consistent definitions of data elements to manage cloud-based information services.

34 Part I: Introducing Cloud Computing

Security in the cloud

Companies planning to use cloud services must be assured of tight, well-defined security services. Chapter 15 details security.

Many levels of security are required within a cloud environment:

- ✓ **Identity management:** For example, so that any application service or even hardware component can be authorized on a personal or group role basis.
- ✓ **Access control:** There also needs to be the right level of access control within the cloud environment to protect the security of resources.
- ✓ **Authorization and authentication:** There must be a mechanism so the right people can change applications and data.

A comprehensive security infrastructure must be provided at all levels and types of cloud services. Developers also need tools that allow them to secure the services they design to be delivered in the cloud. Organizations need consistent security across their own data center environments that intersect with a cloud service.

Managing Cloud Resources

In theory, cloud-services-based resources should be no different from the resources in your own environment, except that they live remotely. Ideally, you have a complete view of the resources you use today or may want to use in the future. Although this sounds straightforward, achieving it isn't that easy.

In most cloud environments, the customer is able to access only the services they're entitled to use. Entire applications may be used on a cloud services basis. Development tools are sometimes cloud based. In fact, testing and monitoring environments can be based on the cloud.

How should you, the cloud customer, approach managing cloud resources? Three aspects of cloud resource management apply:

- ✓ IT security
- ✓ Performance management
- ✓ Provisioning

Chapter 3: Getting Inside the Cloud 35

IT security

IT security is a major concern for new adopters of cloud computing. Ideally, you want the IT security in the cloud to integrate seamlessly with the IT security in your own data center.

However, the cloud service provider implements its own IT security procedures

- ✓ To protect customers from external threats
- ✓ To ensure that individual customer environments are isolated from one another

For every type of cloud service, the provider delivers a good deal of the IT security. You may need to understand how the cloud provider handles issues such as patch management and configuration management as the provider upgrades to new tools and new operating systems.

As the customer, you should

- ✓ Understand the IT security software and hardware (firewalls, intrusion detection systems, virtual private networks [VPNs], and secure connections) that the cloud provider has in place.
- ✓ Know how the cloud providers are protecting the overall computing environment.

In the case of Infrastructure as a Service and Platform as a Service, cloud providers need to clarify the kind of IT security it expects the customer to put in place on its own behalf. With Software as a Service, the provider is responsible for all security except for access security — either an identity management system or at least a local access control application — through the customer's own systems.

Performance management

Performance management is all about how your software services run effectively inside your own environment and through the cloud.

If you start to connect software that runs in your own data center directly to software that runs in the cloud, you create a potential bottleneck at the point of connection.

36 Part I: Introducing Cloud Computing

When you move applications or services into the cloud, you change the basic data center network topology and some application configurations (and possibly some interfaces). This means that performance needs to be considered and designed in at the start for every type of cloud service: Infrastructure as a Service, Platform as a Service, and Software as a Service.

Services connected between the cloud and your computing environment can impact performance if they aren't well planned. This is especially likely to be the case if there are data translations or specific protocols to adhere to at the cloud gateway.

As a customer, your ability to directly control the resources will be much lower in the cloud. Therefore,

- ✓ The connection points between various services must be monitored in real time. A breakdown may impact your ability to provide a business process to your customers.
- ✓ There must be expanded bandwidth at connection points.

From the performance perspective, the situation is likely to be much less fragile if systems don't straddle the data center and the cloud.

In reality, many companies combine services in the cloud and services within their own data center. Therefore, monitoring across these environments prevents many problems.

Provisioning

With Software as a Service, a customer expects provisioning (to request a resource for immediate use) of extra services to be immediate, automatic, and effortless. The cloud service provider is responsible for maintaining an agreed-on level of service and provisions resources accordingly.

The situation is similar with Platform as a Service or Infrastructure as a Service, but you may need to directly request additional resources — because in both cases you're directly managing the cloud resources instead of having them managed on your behalf.

The normal situation in a data center is that software workloads vary throughout the day, week, month, and year. So the data center has to be built for the maximum possible workload, with a little bit of extra capacity thrown in to cover unexpectedly high peaks. One of the immediate attractions of Infrastructure as a Service is that a data center could move its volatile workloads into the cloud and pay for additional resources on demand. In other words, hardware use in the data center is much more efficient.

Service management

Service management in this context covers all the data center operations activities. This broad discipline considers the necessary techniques and tools for managing services by both cloud providers and the internal data center managers across these environments:

- ✓ Physical
- ✓ IT
- ✓ Virtual

Service management encompasses many different disciplines, including

- ✓ Configuration management
- ✓ Asset management
- ✓ Network management
- ✓ Capacity planning
- ✓ Service desk
- ✓ Root cause analysis
- ✓ Workload management
- ✓ Patch and update management

The reality is that the cloud itself is a service management platform. Therefore, well-designed cloud service portfolios include a tight integration of the core service management capabilities and well-defined interfaces.

Untangling Software Dependencies

Perhaps the biggest problem that organizations face when they consider moving applications systems or whole collections of systems into the cloud is the issue of software dependencies. This is important because the cloud uses a service-oriented architecture approach where these services are loosely coupled (loosely linked) together. How do you deal with the dependencies between services? See Chapter 19 for more on loose coupling and service orientation.

In theory, all the applications running your data center share infrastructure and data. However, as companies begin migrating applications or capabilities (such as data or storage) into the cloud, these applications need to be designed to support technical independence.

38 Part I: Introducing Cloud Computing

Therefore, before any part of the data center moves to the cloud, it needs to be separated from dependencies that already exist. For a gradual move into the cloud, the full inventory of applications and systems needs to be considered in terms of which ones *must* — or at least *should* — move together. The interdependencies determine this, whatever their perspective:

- ✓ Hardware perspective (for example, a specific clustering of processors is required)
- ✓ Platform perspective (say, the provider must use a given OS and a given middleware product)
- ✓ Software perspective (maybe the specific services need to be closely coupled with related applications)
- ✓ From any mixture of these

Removing the dependencies among various applications and systems helps an organization evolve the data center into a more flexible, modular environment. Establishing this architectural approach, in effect, creates containers of functionality within the data center that ease the transition to the cloud.

Chapter 4

Developing Your Cloud Strategy

In This Chapter

- Knowing your company's strategy
- Assessing your environment
- Doing the math
- Weighing the risks

Many companies think that the cloud has the potential to dramatically

reduce the costs of managing their technology infrastructure. The situation isn't black and white. In some situations, cloud-based services are the right solution at the right time at the right price. In other circumstances, the cloud as a platform needs more investigation before applying it to a business problem. It is tempting to simply start using a variety of cloud services in response to an immediate problem. However, before you get too far down the road, you need to make sure that you have a strategy and road map for how and when you are going to use cloud services within your organization.

You need a strategy. Here, we list five key areas that should be part of your planning:

- ✓ When and how should you use a public, private, and hybrid cloud service?
- ✓ What is your company's strategy for managing capital and operational expenses over time?
- ✓ How do you plan to achieve the right level of service across the cloud and the data center?

- ✓ What are the rules and regulations that your cloud provider needs to adhere to, to keep your company safe and in compliance?
- ✓ How are you planning to control the data as it moves out of your data center into external clouds?

40 Part I: Introducing Cloud Computing

There is no *one* right path or strategy to leveraging cloud services within your business. The decision depends on your data center, your applications, your service portfolio, and your changing business requirements.

Seeing the Many Aspects of Your Cloud Strategy

You have to think about several issues before sending your organization into the cloud. There isn't just one approach. You might choose one or more of these approaches at different times for different reasons.

Consider a few simple examples:

- ✓ **Your company is building a new application that will change the way you sell products online.** You want to stress test this new application before releasing it to customers. Although you have a few extra resources inside your firewall, they aren't extensive enough to demonstrate if the new application will really scale. Using a cloud Infrastructure as a Service enables you to test the application effectively.
- ✓ **Your company has run its own email internally for more than 20 years.** It takes up a lot of space in the data center and requires a staff of ten people. Money is tight and the CIO must cut staff and capital expenses. The CIO finds a Software as a Service platform that can run the corporate email for a fraction of the cost of running email internally. Your company makes the move and the savings are dramatic.
- ✓ **Your company is building a new but highly experimental application that might transform its business model.** It might not be worth spending a lot of money on software and hardware upfront. In fact, if the project succeeds, the new application may be deployed in the cloud (and not within your company's own data center). Therefore, the company uses a *Platform as a Service (PaaS)* that includes its own well-designed and fully vetted development environment, new generation tools, and interfaces that allow it to connect to many different environments. No need to pretest all the components provided by the PaaS vendor — they're well designed and have been tested. The new application built on this platform is completed in record time and deployed to a test group of customers directly from the cloud service.
- ✓ **Your company has started using a third-party SaaS solution for its customer-management application.** It has successfully replaced the on-premise customer-relationship management package that you've been running in the data center for years. Now your company wonders what else could be moved out of the data center into the cloud. How about the mainframe transaction processing system that handles all orders

worth more than \$1 million? After some investigation, you realize that

Chapter 4: Developing Your Cloud Strategy 41

because the system is only used by a few individuals in the company and the information needs to be carefully governed, the cloud isn't a good choice.

- ✓ **Your CIO has seen some new software that could solve a serious problem, but you aren't convinced that the solution is right.** Instead of buying a license, your company decides to use it as a service. After six months, it proves valuable. The software company offers you the opportunity to use the Software as a Service or on premise.

As you can see, planning your cloud strategy has many different dimensions — maybe more than what you might have thought about in the past. You need a road map to think about how a cloud strategy can be used to support your company's business goals.

Questioning Your Company's Strategy

Deciding to investigate cloud computing is primarily a business decision. Therefore, start by evaluating these things:

- ✓ Your company's business strategy
 - ✓ The role that your technology infrastructure plays in that strategy

Ask these questions:

- ✓ What type of business am I in?
- ✓ Is my industry highly regulated?
- ✓ How do I benefit my customers?
 - ✓ How much value am I getting from my existing data center operations?
- ✓ What are my company's short-term and long-term goals?
- ✓ Am I under pressure to reduce capital expenses?
- ✓ Am I planning to develop new, untested technology-based offerings over the next 18–24 months?
- ✓ Am I looking to acquire complementary companies?
 - ✓ How are my competitors dealing with their technology strategy?
 - ✓ Are my competitors able to move more quickly than my company?
- ✓ What are my strategic advantages?

After you understand the company strategy, you can develop your own strategic plan.

Assessing Where You Are Today

The blessing and the curse of cloud computing is that it is actually quite easy to get started. In many cases, you start with the cloud as a technical approach first and then move to a strategic approach over time. One of the benefits of cloud computing is that for very low cost you can try out a service, find out how it works, and how it might benefit your company. You simply sign yourself up for a service.

And that might be a fine starting point. However, make sure that you're thinking strategically, not just about one or a series of cloud services; investigate how these services fit into your overall computing strategy. You need to ask yourself the following questions.

How tangled is my computing environment?

Whether you're a small or large company, you probably have a lot of applications. Some applications may have been developed internally, whereas others may be packaged software. What makes a computing environment complicated? There could be hundreds of these applications with thousands of dependencies that are difficult to untangle. For example, your company might have a single mortgage-management system that depends on ten other accounting and customer-management systems. Your company might rely on external services from a third party (a provider of payment services, for example). Indeed, most data centers have evolved over time into a complex, tangled web of applications, servers, and networks.

What's my data center environment?

When you consider peeling off some applications and moving them to the cloud, think about the impact on your overall business. Many computing environments aren't set up as a series of well-defined services.

Analyze your computing environment's current state. This type of analysis gives you clues to possible cost savings. Instead of moving applications directly to the cloud, you might actually have to start by simplifying your internal computing environment.

Chapter 4: Developing Your Cloud Strategy 43

Take a hard look at your data center:

- ✓ Is the architecture consistent or does each application have its own architecture?
- ✓ Are there common business services used by multiple applications?
- ✓ Are these common business services self-contained (or do they depend on other applications and services in your environment)?

- ✓ Has your organization implemented workload management?

The more service enabled your computing environment becomes, the better prepared your organization will be to take advantage of various cloud services. It's much harder to move components into the cloud if you can't separate applications or business services from your data center services.

What data supports my strategy?

Consider the information running in your data center. Before moving any data into the cloud, you need to think about

- ✓ Privacy and compliance issues
- ✓ Security issues
- ✓ Data management issues specific to your company

For example, if you're a financial services organization that moves your email into the cloud, email must be archived. After archiving the email, it must be easily accessible to meet regulatory considerations. Another example: A multinational company, before moving data into the cloud for marketing analysis, checks the laws regarding data access by different governments, cross-country data flow, and so on.

Data management and security are so important that we devote two separate chapters to each topic: See Chapters 8 and 15. In fact, your data may require reevaluation of a public cloud model to a private or hybrid model. Chapter 9 details private and hybrid clouds.

It might seem a lot of work to go through the exercise of understanding your current environment. It's crucial to helping you pick the right cloud strategy for your organization.

- ✓ You may think the best starting point is to implement Infrastructure as a Service to add incremental storage capacity to support a new business initiative.

44 Part I: Introducing Cloud Computing

- ✓ You might decide to use Platform as a Service to limit the capital expenses needed to develop a new application.
- ✓ Another starting point might be to add Software as a Service to analyze what the market is saying about your products and any possible acquisition targets.
- ✓ Some organizations might have the need for a Business Process as a Service (such as a supply chain service on demand) that could support testing a new line of business.

Assessing Your Expense Structure

One of the most important tasks when preparing for the cloud: Assessing your

cost structure (for example, how much you're spending on supporting existing hardware, software, networking services). How can you determine the cost savings if you don't know what you're spending today? Also take potential future costs into account.

Things may get fuzzy. You may sometimes want to use business services offered by cloud application vendors. You may want to build some internal service oriented architecture-based services that can live inside a cloud environment. In some situations, it may save money to move a service such as email, software testing, or storage to a cloud, because the costs of performing the service internally are so much higher. In other situations, the costs for implementing a key application in the cloud may be much more expensive than running it internally.

Chapter 21 explains more about cloud economics.

Checking Up on Rules and Governances

We recommend assessing your current IT and business governance situation as you develop your cloud strategy. In some cases, governance and compliance prohibit certain types of information from leaving the organization's internal environment. How good is your internal security today? If you're considering a cloud service provider, you need to be confident that the company can support your security and governance needs with oversight and accountability. Examine the reports and documentation to support your oversight requirements. Talk to the provider's other customers to see how well it meets its customers governance requirements.

Chapter 4: Developing Your Cloud Strategy 45

For example, you may want to leverage a third-party credit checking service from the cloud. How well constructed is it? Does it conform to your company's business rules?

Aside from security and privacy issues, you have a number of legal issues to consider as well. For example, what happens to your application and data if the cloud provider goes out of business? Who's liable for lost information? Does the provider guarantee uptime? What recourse do you have if the service level agreement isn't met?

Chapter 16 details governance issues.

Developing a Road Map

You must consider many things before developing a road map:

- ✓ The efficiency and effectiveness of your current data center
- ✓ Costs
- ✓ Risks
- ✓ Your organizational readiness

After you understand the issues and gaps, you can start designing your cloud plan — the road map that outlines the following:

- ✓ What are the services that you need to support your business growth?
- ✓ How you will roll them out?
- ✓ When you will roll them out (or *in*, as it were)?

Don't try to do everything at once with your cloud strategy. It probably makes sense to roll out these services gradually so you can see the benefits and get buy-in throughout your organization. Plus, starting cloud services step by step can help you react quickly to business needs.

Even if you figure out all the technical requirements for leveraging the cloud as part of your strategy, you still have to plan to communicate the action plan to the business and the IT communities. Some people might consider the cloud a threat because it will remove some tasks from the IT department. Business management will want to know that they have control over important business data. For more details on your strategy action plan see Chapter 22.

46 Part I: Introducing Cloud Computing

You need to understand how your vendors track performance and security. Don't simply take their word for it and assume that everything is perfectly fine. Even if the cloud vendor provides you with a slick dashboard, you should have your own means of monitoring your content. You're turning over some key responsibility to a cloud provider, but the buck still stops with your organization. Plan carefully for controlling your assets in the cloud. Chapter 20 talks more about management from a cloud customer perspective.

Part II

Understanding the Nature of the Cloud

The 5th Wave

By Rich Tennant



"We're still working out the kinks in our cloud computing environment."

In this part . . .

What's inside the cloud? In this part, we examine a

highly scaled computing environment. Because that environment is front and center, we look at the technical foundation for this model, including workloads and data services.

Chapter 5

Seeing the Advantages of the

Highly Scaled Data Center

In This Chapter

- ▶ Modeling a data center
- ▶ Location, location, location
- ▶ Powering things up
- ▶ Cooling things off

As we discuss in Chapter 1, many company managers are demanding that IT management transform their data centers into platforms that can scale easily and effectively. Other managers are looking at the cloud platform as a way to eliminate the high costs of running traditional data centers.

If you're tasked with planning your cloud strategy, how do you do what's best for your organization? At first glance, it might seem obvious: Simply find a cloud services provider, analyze how much it charges for the services you need, and compare it to the costs of your own data center. It isn't that simple.

- ✓ It's unlikely that everything you do in your data center will be available as a cloud service.
- ✓ Even if it is, it might not meet your specific needs.

Ultimately, cloud services are attractive because the cost is likely to be far lower than providing the same service from your traditional data center, so we think it will help if you understand *why* cloud data center costs are lower. This economic factor applies to clouds whether they're private or public.

50 Part II: Understanding the Nature of the Cloud

In fact, the cloud data center has two aspects:

- ✓ The costs of things that don't depend directly on technology
- ✓ The costs of things that do

In this chapter, we take an in-depth look at the things that don't depend on technology and explain why the cloud data center has a significant cost advantage.

Comparing Financial Damage:

Traditional versus Cloud

How much does a data center cost to run? It depends on these things:

- ✓ **How big it is.** How many virtual servers? Is the data center massive? How much square footage; how many servers? Does it cost \$5 million a year to run?
- ✓ **Where it is.** How much does office space cost. What about cost of staff? Is the data center close to inexpensive power sources?
- ✓ **What it's doing.** Does the data center protect sensitive data? What is its kind of business? What level of compliance must it adhere to?

Clearly, you have many ways to look at the situation.

Traditional data center

Although each data center is a little different, the average cost per year to operate a large data center is usually between \$10 million to \$25 million.

Stranger than fiction

We didn't make up the \$10 million to \$25 million number. In 2008, *BusinessWeek Magazine* published an article called "Computing Heads for the Clouds," by Rachael King ([http://](http://images.businessweek.com/ss/08/08/0804_cloudcomputing/1.htm)

images.businessweek.com/ss/08/08/0804_cloudcomputing/1.htm). The magazine surveyed 11 different large data centers throughout the United States.

Chapter 5: Seeing the Advantages of the Highly Scaled Data Center 51

Where's the bulk of the money going? This might surprise you.

- ✓ **42 percent:** Hardware, software, disaster recovery arrangements, uninterrupted power supplies, and networking. (Costs are spread over time, *amortized*, because they are a combination of capital expenditures and regular payments.)
- ✓ **58 percent:** Heating, air conditioning, property and sales taxes, and labor costs. (In fact, as much as 40 percent of annual costs are labor alone.)

The reality of the traditional data center is further complicated because most of the costs maintain existing (and sometimes aging) applications and infrastructure. Some estimates show 80 percent of spending on maintenance.

Before you conclude that you need to throw out the data center and just move to the cloud, know the nature of the applications and the workloads at the core of data centers:

- ✓ Most data centers run a lot of different applications and have a wide variety of workloads.

- ✓ Many of the most important applications running in data centers are actually used by only a relatively few employees. For example, transaction management applications (which are critical to a company's relationship to customers and suppliers) might only be used by a few employees.
- ✓ Some applications that run on older systems are taken off the market (no longer sold) but are still necessary for business.

Because of the nature of these applications, it probably *wouldn't* be cost effective to move these environments to the cloud.

Cloud data center

In this case *cloud data centers* means data centers with 10,000 or more servers on site, all devoted to running very few applications that are built with consistent infrastructure components (such as racks, hardware, OS, networking, and so on).

What's the key difference in the cost structure of a traditional data center and a cloud data center? One of the most important factors is that cloud data centers aren't remodeled traditional data centers.

52 Part II: Understanding the Nature of the Cloud

Cloud data centers are

- ✓ Constructed for a different purpose.
- ✓ Created at a different time than the traditional data center.
- ✓ Built to a different scale.
- ✓ Not constrained by the same limitations.
- ✓ Perform different workloads than traditional data centers.

Because of this design approach, the economics of a cloud data center are significantly different.

To create a basis for analyzing this, we used figures on the costs of creating a cloud data center described in a Microsoft paper titled "The Cost of a Cloud: Research Problems in Data Center Networks" by Albert Greenberg, James Hamilton, David A. Maltz, and Parveen Patel.

We took estimates for how much it cost to build a cloud data center and looked at three cost factors:

- ✓ **Labor** costs were 6 percent of the total costs of operating the cloud data center.
- ✓ **Power distribution and cooling** were 20 percent.
- ✓ **Computing** costs were 48 percent.

Of course, the cloud data center has some different costs than the traditional data center (such as buying land and construction).

This explanation of costs is designed to give you an idea of where the difference between the traditional data center and the cloud data center are. The upfront costs in constructing cloud data centers are actually spread across hundreds of thousands of individual users. Therefore, after they're constructed, these cloud data centers are well positioned to be profitable because they support so many customers with a large number of servers executing a single application.

Scaling the Cloud

From the provider's point of view, the whole point of cloud computing is to achieve economies of scale by managing a very large pool of computing resources in a highly economic and efficient fashion.

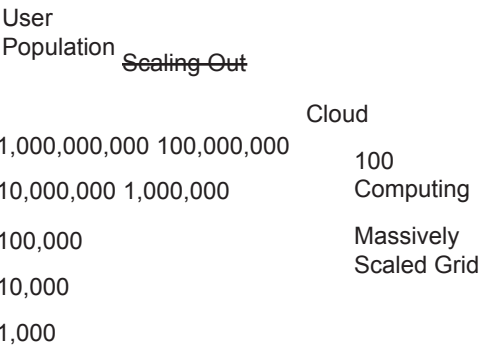
Chapter 5: Seeing the Advantages of the Highly Scaled Data Center 53

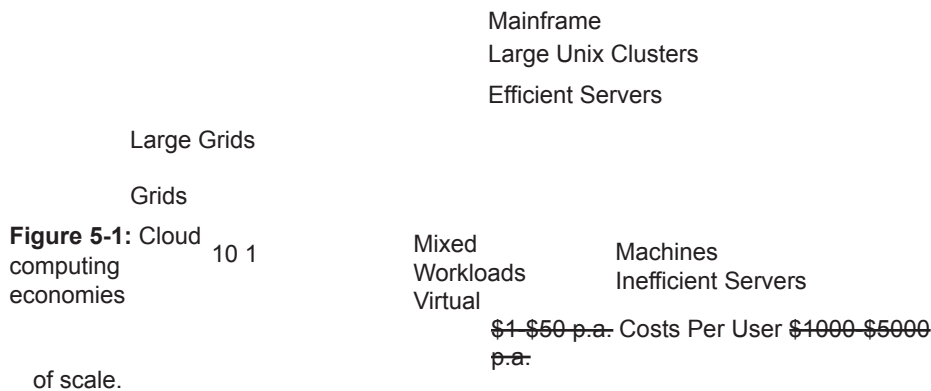
A picture makes it a little clearer. Figure 5-1 shows a graph of the cost per user of running just one software application using different kinds of computer resources; this is charted against the number of users. We need to emphasize that we're talking about just *one* application — not even two or three. In Figure 5-1, that one application runs in different computing environments, starting with inefficient dedicated servers all the way up to massively scaled grids.

An important point to note is that the Y-axis of user populations is logarithmic. That means that the curve is much less steep than if we drew it on a proportional scale of equal steps. If we drew it on a proportional scale, we'd need miles of paper.

We deliberately *didn't* put units on the X-axis. Instead, note the following:

- ✓ One end of the X-axis shows data center costs between \$1–\$50 per user per annum. That reflects, for example, the prices that Google charges for Google Apps or even the cost of providing free email (from Google, Microsoft, or Yahoo, which is paid for by ads). The cost per user is extremely low.
- ✓ The other end of the X-axis shows data center costs between \$1,000–\$5,000 per user per annum. That might be the cost of, for example, providing a print server that's almost always idle.





54 Part II: Understanding the Nature of the Cloud

Basically, on the left in Figure 5-1 you have very efficient use of computer resources and, on the right, very *inefficient* use of resources.

Points on the line indicate the kind of computing resources that serve specific group sizes:

- ✓ **Inefficient servers:** This is a 1:1 user-to-server ratio (or close to 1). The cost of managing a single server in a data center will be thousands of dollars per year and this is as expensive as computing ever gets per user.
- ✓ **Virtual machines:** Applications and user numbers that can't use a whole server get *virtualized* (split among several virtual servers). This is efficient (making better use of underused servers), but also inefficient (virtualization requires significant overhead, as does running the multiple guest operating systems).
- ✓ **Efficient servers (and small clusters):** User populations from the hundreds to 1,000 can be served reasonably efficiently with a single or multiple servers if there's only one application being run on a server; servers can be highly efficient, yielding a relatively low cost per user.
- ✓ **Mainframe and large Unix clusters:** They're shown separately on the grid only for the sake of space. Both can handle very large database applications from thousands to tens of thousands of users.
- ✓ **Grids:** From the hundreds of thousands to a million users, you're in the area where *Software as a Service (SaaS)* vendors such as Salesforce.com operate. Business applications offered by SaaS vendors present a thorny scaling problem because it's a transactional database application. The main Salesforce.com CRM application runs on a grid of about 1,000 computers.
- ✓ **Large grids:** Concurrent users above one million. Still a very heavy workload and only possible via a *scale-out* (which lets a single workload expand by using more of the identical inexpensive resources) approach with a grid. Twitter and Linked-In are examples.
- ✓ **Massively scaled grid:** This is for user populations in the tens of millions. Example: Each query on Google search is resolved by a purpose-built grid of up to 1,000 servers; Google routes queries to many such grids. Yahoo also has a massively scaled-out email system. It caters to more than 260 million users, of which tens of millions must be active at a time.

The dotted box in Figure 5-1 indicates the *traditional* domain and kinds of resources of corporate computing. The same servers used in corporate environments could be used just as easily in scaled-out arrangements, where workloads aren't at all mixed. The reduction in per-user costs doesn't, at

Chapter 5: Seeing the Advantages of the Highly Scaled Data Center 55

the moment, come from using different computer equipment or different operating systems: It comes from running a small number (or even just one) workload and scaling it up as much as possible. That's how cloud computing reduces costs dramatically.

No corporation that runs a mixed workload is ever going to achieve cloud computing's economies of scale.

But how do massively scaled data centers manage to get their per-user costs so very low? This becomes clear when you read about each area of data costs in Chapter 21.

Comparing Traditional and Cloud Data Center Costs

Before reading how to reduce data center costs, reread the *traditional* IT costs statistics:

- ✓ Portion of IT budget used to maintain and run existing systems: 70–80 percent
- ✓ Portion of IT budget used to build and implement new capabilities: 20–30 percent

Compare traditional and cloud data centers in Table 5-1.

Table 5-1 A Comparison of Corporate and Cloud Data Centers

<i>Traditional Corporate Data Center</i>	<i>Cloud Data Center</i>
Thousands of different applications ^{one)}	Few applications (maybe even just
Mixed hardware environment	Homogeneous hardware environment
Multiple management tools	Standardized management tools
Frequent application patching and updating	Minimal application patching and updating
Complex workloads	Simple workloads
Multiple software architectures	Single standard software architecture

Looking at the table, it becomes clear that the cloud data center is much simpler to organize and operate and, because it is simple, it scales well. In other words, the larger you make it, the lower the costs per user are. In the next section, we examine some of these costs and see where the efficiencies arise.

Examining labor costs and productivity

Labor costs depend on several things:

- ✓ **Technology managing the data center:** Even improving that technology in a traditional corporate setting may reduce the cost of labor only a small amount.
- ✓ **In what environment someone works:** The labor cost per person is likely to be equivalent regardless of the data center type; the skills requirement is the same. But that person's *productivity* varies depending on the environment. Operating the scaled cloud data center is much simpler.

The impact of this set of differences on labor costs is dramatic. Corporate data centers usually have a ratio of operational staff to servers of around 1 person to 65 servers. In cloud data centers, that ratio is more like 1 person to 850 servers, and we've even come across better ratios than that. This is a 10-to-1 improvement in the productivity of labor (or possibly more — maybe going as high as 20 to 1).

Wondering where you are

The traditional setup's 58 percent costs depend a lot on location:

- ✓ Electricity fees
- ✓ Local taxes
- ✓ Labor costs

Compare a data center in North Carolina with one in New York (keeping in mind no two data centers have the same software workloads). Better to consider technology costs separately and see where economies arise, which we do in the following sections.

Chapter 5: Seeing the Advantages of the Highly Scaled Data Center 57

Electric power

Computers have been using more electricity in recent years and, at 7 percent of corporate data centers' costs (including heating and cooling), the cost is significant. Cloud data centers use even more: Electricity costs hover around 12 percent.

Cloud data centers can do the following:

- ✓ **Put the data center where the cheap power is.** Electricity fluctuates in price from year to year and costs are difficult to control.

✓ **Negotiate a discounted power contract with its power company.**

Cloud data centers, by their level of usage, fall into the least expensive category.

If a cloud data center is contemplating building a data center, it can negotiate a long-term deal for an even deeper discount than industrial usage gives them. Put the data center very close to the power plant and bargain for a lower cost supply based on these points:

- Distance from the power station (because less electrical power is lost in transit).
- Minimal power interruption from electrical storms (if you have a private circuit direct to the power station).

Outsourcing

Because power is so critical to the cloud data center, organizations have to consider the availability and cost of energy sources as they would any primary data center resource.

Electricity sources include the following:

- ✓ Hydroelectric is generally expensive when it has to travel far to customers, but otherwise it's usually cheap and can be the ideal source of power for a data center.
- ✓ Oil prices change, which can cause cost fluctuation.
- ✓ Liquefied natural gas (LNG) suffers from the same changing fuel prices as oil.
- ✓ Coal is more stable in price, but not green.
- ✓ Nuclear is inexpensive to run but expensive to build and gain approval

58 Part II: Understanding the Nature of the Cloud

Other location costs

Other location related costs when building a new data center include the following:

- ✓ **Land costs:** The days of siting data centers in skyscrapers in Manhattan are over. Better to use cheap land with low property taxes. There are exceptions, of course. For example, in algorithmic financial trading, latency lost due to networking (communications) distance directly impacts revenue.
- ✓ **Building costs:** A designed-entirely-as-a-data-center building is a must.
 - *Heat management* is the overriding priority, so building *out* almost certainly makes more sense than building *up*. Cool geographical areas may make more sense than hotter ones.
 - *Safety* is another important consideration. Data centers need to be electrically safe, secure, and fireproof.
- ✓ **Staff:** Although staff costs are very low for the cloud data center, as a percentage of the whole, location in areas (or even countries) where staff costs are low can further reduce staff costs.
- ✓ **Investment incentives and taxation:** Many areas of the world, including states in the United States, welcome inward investment and help finance it with very generous tax exemptions and cash incentives. Take advantage

tage of these opportunities when you find them.

In the next chapter, we examine technology costs, which also favor the cloud data center in many ways. The simple fact is that data centers as they exist now, in the enterprise, are a cottage industry that's going to change in the coming years by the mass-production efficiencies of cloud data centers.

Chapter 6

Exploring the Technical Foundation for Scaling Computer Systems

In This Chapter

- ▶ Comparing traditional data centers to clouds
- ▶ Achieving economies of scale
- ▶ Saving money via the bottom line

In Chapter 5, we contrast the non-technology operational costs of the traditional data center with those of the cloud data center (electricity, cooling, space, and so on). In this chapter, we contrast technology costs between the traditional data center and the cloud data center.

We divided into four areas the places where IT spends money:

- ✓ **Hardware**, including servers, storage, and so on
- ✓ **A power supply** for those systems and how to keep them from overheating
- ✓ **Networking and communications equipment** so the systems can interoperate
- ✓ **Electricity** to support the overall data center

Some elements are more expensive than others. In Chapter 5, we look at two reports that detail the costs of running traditional and cloud data centers. Using this same set of numbers, we calculated the costs of the areas. The results are quite interesting. The greatest expense in the traditional data center is server and storage hardware, which accounts for 36 percent of the amortized costs. The second biggest expense? Power distribution and cooling. Amortized over a year, power and cooling are 20 percent of the total

expenses. Both networking and electricity each add 12 percent to the total expense number per year. Add hardware and its supporting power and cooling, and you have 56 percent of the technology related costs.

We discuss electricity costs in the previous chapter, but only from the perspective of arranging for an inexpensive supply. In this chapter, we take on the issue of using that electricity efficiently.

Server-ing Up Some Hardware

Although we'd like to tell you that costs are static, clearly they aren't. Costs for your data center hardware will vary dramatically depending on the type of workloads you support.

Data storage is an excellent example of this variation. If a data center is feeding video to the Internet from a vast video library (like YouTube does) the storage requirements are huge. However, storing short text messages (as

Twitter does) doesn't require a lot of space. Indeed, Twitter doesn't even store its billions of messages indefinitely. The YouTube library, on the other hand, just keeps on growing.

Tradition! versus clouds

What does this mean when you look at the differences in the costs of hardware between the traditional data center and the cloud data center? Look at a snapshot of each:

- ✓ **Tradition:** In a traditional data center, IT management has a structured process for purchasing hardware. Each year they talk to business units, determine what new applications to add or expand, and work with vendors on the procurement process. In addition, most IT organizations refresh their hardware on a regular basis to make sure that things run smoothly and old systems are retired before they cause problems.
- ✓ **Cloud:** When a business is creating a cloud data center (either a private one inside the firewall or a service provider) the process of procuring systems is very different. Because the cloud supports very different workloads, IT management doesn't buy traditional hardware. Rather, IT management might go directly to an engineering company that designs the system boards and networking switches for them, and then take the contract to a manufacturer to have them build the precise hardware that they want.